```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

***********************************************************************
*
*        EXPERIMENTAL pending further PoP definition
*
*        Zvector E6 instruction tests for VRR-c encoded:
*
*        E675  VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
*
*        and partial testing of
*
*        E656 VCLFNH   - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
*        E65E VCLFNL   - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
*
*        during cross check tests for VCRNF
*
*        James Wekel August 2024
***********************************************************************


***********************************************************************
*
*        basic instruction tests
*
***********************************************************************
*  This program tests EXPERIMENTAL functioning of the z/arch E6 VRR-c
*  Neural-network-processing-assist facility vector instructions.
*  These test are EXPERIMENTAL pending further PoP definition of
*  NNP-data-type-1.
*
*  If requested and if VXC == 0 after test instruction execution,
*  a cross check test is performed. A cross check uses the result
*  of the instruction test to recreate the test source.
*
*  Exceptions (including trapable IEEE exceptions) are not tested.
*
*  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
*  obvious coding errors.  None of the tests are thorough.  They are
*  NOT designed to test all aspects of any of the instructions.
*
***********************************************************************
*
*   *Testcase zvector-e6-21-VCRNF
*   *
*   *        EXPERIMENTAL pending further PoP definition
*   *
*   *        Zvector E6 instruction tests for VRR-c encoded:
*   *
*   *        E675 VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
*   *
*   *        # -------------------------------------------------------
*   *        #  This tests only the basic function of the instruction.
*   *        #  Exceptions are NOT tested.
*   *        # -------------------------------------------------------
*   *
*   mainsize    2
*   numcpu      1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                    57 *   sysclear
                                                    58 *   archlvl     z/Arch
                                                    59 *
                                                    60 *   loadcore    "$(testpath)/zvector-e6-21-VCRNF.core" 0x0
                                                    61 *
                                                    62 *   diag8cmd    enable    # (needed for messages to Hercules console)
                                                    63 *   runtest 5
                                                    64 *   diag8cmd    disable   # (reset back to default)
                                                    65 *
                                                    66 *   *Done
                                                    67 *
                                                    68 ***********************************************************************
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                    70 ***********************************************************************
                                                    71 *        FCHECK Macro - Is a Facility Bit set?
                                                    72 *
                                                    73 *        If the facility bit is NOT set, an message is issued and
                                                    74 *        the test is skipped.
                                                    75 *
                                                    76 *        Fcheck uses R0, R1 and R2
                                                    77 *
                                                    78 * eg.    FCHECK 134,'vector-packed-decimal'
                                                    79 ***********************************************************************
                                                    80          MACRO
                                                    81          FCHECK &BITNO,&NOTSETMSG
                                                    82 .*                         &BITNO : facility bit number to check
                                                    83 .*                         &NOTSETMSG : 'facility name'
                                                    84          LCLA  &FBBYTE           Facility bit in Byte
                                                    85          LCLA  &FBBIT            Facility bit within Byte
                                                    86
                                                    87          LCLA  &L(8)
                                                    88 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                    89
                                                    90 &FBBYTE  SETA  &BITNO/8
                                                    91 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                    92 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                    93
                                                    94          B     X&SYSNDX
                                                    95 *                                     Fcheck data area
                                                    96 *                                     skip messgae
                                                    97 SKT&SYSNDX DC  C'          Skipping tests: '
                                                    98          DC    C&NOTSETMSG
                                                    99          DC    C' facility (bit &BITNO) is not installed.'
                                                   100 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                   101 *                                     facility bits
                                                   102          DS    FD                     gap
                                                   103 FB&SYSNDX DS   4FD
                                                   104          DS    FD                     gap
                                                   105 *
                                                   106 X&SYSNDX EQU *
                                                   107          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                   108          STFLE FB&SYSNDX              get facility bits
                                                   109
                                                   110          XGR   R0,R0
                                                   111          IC    R0,FB&SYSNDX+&FBBYTE    get fbit byte
                                                   112          N     R0,=F'&FBBIT'           is bit set?
                                                   113          BNZ   XC&SYSNDX
                                                   114 *
                                                   115 * facility bit not set, issue message and exit
                                                   116 *
                                                   117          LA    R0,SKL&SYSNDX           message length
                                                   118          LA    R1,SKT&SYSNDX           message address
                                                   119          BAL   R2,MSG
                                                   120
                                                   121          B     EOJ
                                                   122 XC&SYSNDX EQU *
                                                   123          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   125 ***********************************************************************
                                                   126 *        Low core PSWs
                                                   127 ***********************************************************************

                              00000000  00002D83   129 ZVE6TST  START 0
00000000                      00000000             130          USING ZVE6TST,R0            Low core addressability
                                                   131
                              00000140  00000000   132 SVOLDPSW EQU   ZVE6TST+X'140'        z/Arch Supervisor call old PSW

00000000                      00000000  000001A0   134          ORG   ZVE6TST+X'1A0'        z/Architecture RESTART PSW
000001A0  00000001 80000000                        135          DC    X'0000000180000000'
000001A8  00000000 00000200                        136          DC    AD(BEGIN)

000001B0                      000001B0  000001D0   138          ORG   ZVE6TST+X'1D0'        z/Architecture PROGRAM CHECK PSW
000001D0  00020001 80000000                        139          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                        140          DC    AD(X'DEAD')

000001E0                      000001E0  00000200   142          ORG   ZVE6TST+X'200'        Start of actual test program...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   144
                                                   145 *******************************************************************
                                                   146 *               The actual "ZVE6TST" program itself...
                                                   147 *******************************************************************
                                                   148 *
                                                   149 *  Architecture Mode: z/Arch
                                                   150 *  Register Usage:
                                                   151 *
                                                   152 *   R0       (work)
                                                   153 *   R1-4     (work)
                                                   154 *   R5       Testing control table - current test base
                                                   155 *   R6       (work)
                                                   156 *   R7       msg return
                                                   157 *   R8       First base register
                                                   158 *   R9       Second base register
                                                   159 *   R10      Third base register
                                                   160 *   R11      E6TEST call return
                                                   161 *   R12      E6TESTS register
                                                   162 *   R13      (work)
                                                   163 *   R14      Subroutine call
                                                   164 *   R15      Secondary Subroutine call or work
                                                   165 *
                                                   166 *******************************************************************

00000200                     00000200              168          USING  BEGIN,R8                 FIRST Base Register
00000200                     00001200              169          USING  BEGIN+4096,R9            SECOND Base Register
00000200                     00002200              170          USING  BEGIN+8192,R10           THIRD Base Register
                                                   171
00000200  0580                                     172 BEGIN    BALR  R8,0                      Initialize FIRST base register
00000202  0680                                     173          BCTR  R8,0                      Initialize FIRST base register
00000204  0680                                     174          BCTR  R8,0                      Initialize FIRST base register
                                                   175
00000206  4190 8800                     00000800   176          LA    R9,2048(,R8)              Initialize SECOND base register
0000020A  4190 9800                     00000800   177          LA    R9,2048(,R9)              Initialize SECOND base register
                                                   178
0000020E  41A0 9800                     00000800   179          LA    R10,2048(,R9)             Initialize THIRD base register
00000212  41A0 A800                     00000800   180          LA    R10,2048(,R10)            Initialize THIRD base register
                                                   181
00000216  B600 8524                     00000724   182          STCTL R0,R0,CTLR0               Store CR0 to enable AFP
0000021A  9604 8525                     00000725   183          OI    CTLR0+1,X'04'             Turn on AFP bit
0000021E  9602 8525                     00000725   184          OI    CTLR0+1,X'02'             Turn on Vector bit
00000222  B700 8524                     00000724   185          LCTL  R0,R0,CTLR0               Reload updated CR0
                                                   186
                                                   187 *******************************************************************
                                                   188 * Is Neural-network-processing-assist facility 2 installed  (bit 165)
                                                   189 *******************************************************************
                                                   190
                                                   191          FCHECK 165,'Neural-network-processing-assist'
00000226  47F0 80C0                     000002C0   192+         B     X0001
                                                   193+*                                        Fcheck data area
                                                   194+*                                        skip messgae
0000022A  40404040 40404040                        195+SKT0001  DC    C'          Skipping tests: '
00000244  D585A499 81936095                        196+         DC    C'Neural-network-processing-assist'
00000264  40868183 899389A3                        197+         DC    C' facility (bit 165) is not installed.'
                             0000005F  00000001    198+SKL0001  EQU   *-SKT0001
                                                   199+*                                        facility bits
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00000290  00000000 00000000                         200+            DS    FD                          gap
00000298  00000000 00000000                         201+FB0001      DS    4FD
000002B8  00000000 00000000                         202+            DS    FD                          gap
                                                    203+*
                              000002C0  00000001    204+X0001       EQU   *
000002C0  4100 0004                     00000004    205+            LA    R0,((X0001-FB0001)/8)-1
000002C4  B2B0 8098                     00000298    206+            STFLE FB0001                      get facility bits
000002C8  B982 0000                                 207+            XGR   R0,R0
000002CC  4300 80AC                     000002AC    208+            IC    R0,FB0001+20                get fbit byte
000002D0  5400 8560                     00000760    209+            N     R0,=F'4'                    is bit set?
000002D4  4770 80E8                     000002E8    210+            BNZ   XC0001
                                                    211+*
                                                    212+* facility bit not set, issue message and exit
                                                    213+*
000002D8  4100 005F                     0000005F    214+            LA    R0,SKL0001                  message length
000002DC  4110 802A                     0000022A    215+            LA    R1,SKT0001                  message address
000002E0  4520 8438                     00000638    216+            BAL   R2,MSG
000002E4  47F0 8508                     00000708    217+            B     EOJ
                              000002E8  00000001    218+XC0001      EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   220 ***********************************************************************
                                                   221 *              Do tests in the E6TESTS table
                                                   222 ***********************************************************************
                                                   223
000002E8  58C0 8564                      00000764  224          L     R12,=A(E6TESTS)        get table of test addresses
                                                   225
                               000002EC  00000001  226 NEXTE6   EQU   *
000002EC  5850 C000                      00000000  227          L     R5,0(0,R12)            get test address
000002F0  1255                                     228          LTR   R5,R5                     have a test?
000002F2  4780 83F4                      000005F4  229          BZ    ENDTEST                      done?
                                                   230
000002F6                       00000000            231          USING E6TEST,R5
                                                   232
000002F6  4800 5004                      00000004  233          LH    R0,TNUM                save current test number
000002FA  5000 8E04                      00001004  234          ST    R0,TESTING             for easy reference
                                                   235
000002FE  58B0 5000                      00000000  236          L     R11,TSUB               get address of test routine
00000302  05BB                                     237          BALR  R11,R11                do test
                                                   238
00000304  45F0 814E                      0000034E  239          BAL   R15,XCHECK
                                                   240 *
                                                   241 * validate FPC first
                                                   242 *
00000308  D500 500A 5041                 00000041  243          CLC   FLG(1),FPC_R+1          expected FPC flags?
0000030E  4780 8116                      00000316  244          BE    CHECKVXC
00000312  4570 8290                      00000490  245          BAL   R7,FPCMSG               no, issue failed message
                                                   246
                               00000316  00000001  247 CHECKVXC EQU   *
00000316  D500 500B 5042                 00000042  248          CLC   VXC(1),FPC_R+2          expected VXC?
0000031C  4780 8124                      00000324  249          BE    CHECKRES
00000320  4570 8302                      00000502  250          BAL   R7,VXCMSG               no, issue failed message
                                                   251
                               00000324  00000001  252 CHECKRES EQU   *
                                                   253
                                                   254 * then validate results, if not inexact
                                                   255
00000324  B982 0011                                256          XGR   R1,R1
00000328  4310 5041                      00000041  257          IC    R1,FPC_R+1             FPC flags
0000032C  5410 8568                      00000768  258          N     R1,=XL4'00000008'      check inexact flag
00000330  1211                                     259          LTR   R1,R1
00000332  4770 8146                      00000346  260          BNZ   DONEXT
                                                   261
00000336  E310 501C 0014                 0000001C  262          LGF   R1,READDR              expected result address
0000033C  D50F 5028 1000       00000028  00000000  263          CLC   V1OUTPUT,0(R1)
00000342  4770 8374                      00000574  264          BNE   FAILMSG                no, issue failed message
                                                   265
                               00000346  00000001  266 DONEXT   EQU   *
00000346  41C0 C004                      00000004  267          LA    R12,4(0,R12)           next test address
0000034A  47F0 80EC                      000002EC  268          B     NEXTE6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   270 *--------------------------------------------------------------------
                                                   271 * cross check that the result can be converted back to the source
                                                   272 * where there were NO exceptions (DXC/VXC = 0)
                                                   273 *--------------------------------------------------------------------
                                 0000034E  00000001  274 XCHECK   EQU   *
0000034E  D207 5050 8548      00000050  00000748  275          MVC   SKIPXC,=CL8'SKIP XC '
00000354  D500 5007 8574      00000007  00000774  276          CLC   XCSKIP,=CL1'S'         skip xcheck requested
0000035A  478F 0000                     00000000  277          BE    0(R15)                   skip = S, so exit
0000035E  9500 5042                     00000042  278          CLI   FPC_R+2,0              Only Xcheck no VXC error
00000362  477F 0000                     00000000  279          BNZ   0(R15)                   some VXC error , so exit
                                                   280 *
                                                   281 *        cross check depends on the instruction
                                                   282 *
00000366  D207 5050 8550      00000050  00000750  283          MVC   SKIPXC,=CL8'        '
0000036C  D507 5010 8558      00000010  00000758  284          CLC   OPNAME,=CL8'VCRNF'
00000372  4780 8180                     00000380  285          BE    XCVCRNF
                                                   286
                                                   287
00000376  D207 5050 8548      00000050  00000748  288          MVC   SKIPXC,=CL8'SKIP XC '
0000037C  07FF                                     289          BR    R15                    return from xcheck
                                                   290
                                                   291 * cross check: VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                   292 *
00000380                                           293 XCVCRNF  DS    0F
00000380  B29D 852C                     0000072C  294          LFPC  FPCMOFF                initialize FPC
                                                   295
00000384  E700 5028 0806                00000028  296          VL    V16,V1OUTPUT
0000038A  E6F0 0000 2456                           297          VCLFNH V15,V16,2,0
00000390  B29C 5088                     00000088  298          STFPC FPC_XC1                save 1st FPC
00000394  E7F0 5060 000E                00000060  299          VST   V15,XCOUTPUT
                                                   300
0000039A  B29D 852C                     0000072C  301          LFPC  FPCMOFF                initialize FPC
                                                   302
0000039E  E700 5028 0806                00000028  303          VL    V16,V1OUTPUT
000003A4  E6F0 0000 245E                           304          VCLFNL V15,V16,2,0
000003AA  B29C 508C                     0000008C  305          STFPC FPC_XC2                save 2nd FPC
000003AE  E7F0 5070 000E                00000070  306          VST   V15,XCOUTPUT+16
                                                   307
000003B4  9500 508A                     0000008A  308          CLI   FPC_XC1+2,0            1st FPC
000003B8  477F 0000                     00000000  309          BNZ   0(R15)                 some VXC error , so exit
                                                   310
000003BC  9500 508E                     0000008E  311          CLI   FPC_XC2+2,0            2nd FPC
000003C0  477F 0000                     00000000  312          BNZ   0(R15)                 some VXC error , so exit
                                                   313
000003C4  E310 500C 0014                0000000C  314          LGF   R1,V2ADDR              expected source address
000003CA  D51F 5060 1000                00000000  315          CLC   XCOUTPUT(32),0(R1)
000003D0  4770 81D6                     000003D6  316          BNE   XCFAILMSG
000003D4  07FF                                     317          BR    R15                    return from xcheck
                                                   318
                                                   319
                                                   320 * xcheck failed message
000003D6                                           321 XCFAILMSG DS   0H
000003D6  5800 856C                     0000076C  322          L     R0,=F'1'               set failed test indicator
000003DA  5000 8E00                     00001000  323          ST    R0,FAILED
                                                   324
000003DE  4820 5004                     00000004  325          LH    R2,TNUM                get test number and convert
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000003E2  4E20 8F90                     00001190   326          CVD   R2,DECNUM
000003E6  D211 8F7A 8F64      0000117A  00001164   327          MVC   PRT3,EDIT
000003EC  DE11 8F7A 8F90      0000117A  00001190   328          ED    PRT3,DECNUM
000003F2  D202 8F23 8F87      00001123  00001187   329          MVC   XCPTNUM(3),PRT3+13     fill in message with test #
                                                   330
000003F8  D207 8F45 5010      00001145  00000010   331          MVC   XCPNAME,OPNAME         fill in message with instruction
                                                   332
000003FE  B982 0022                                333          XGR   R2,R2
00000402  4320 5008                     00000008   334          IC    R2,M4                  get m3 and convert
00000406  4E20 8F90                     00001190   335          CVD   R2,DECNUM
0000040A  D211 8F7A 8F64      0000117A  00001164   336          MVC   PRT3,EDIT
00000410  DE11 8F7A 8F90      0000117A  00001190   337          ED    PRT3,DECNUM
00000416  D201 8F56 8F88      00001156  00001188   338          MVC   XCPM4(2),PRT3+14       fill in message with m3 field
                                                   339 *
0000041C  B982 0022                                340          XGR   R2,R2
00000420  4320 5009                     00000009   341          IC    R2,M5                  get m4 and convert
00000424  4E20 8F90                     00001190   342          CVD   R2,DECNUM
00000428  D211 8F7A 8F64      0000117A  00001164   343          MVC   PRT3,EDIT
0000042E  DE11 8F7A 8F90      0000117A  00001190   344          ED    PRT3,DECNUM
00000434  D201 8F61 8F88      00001161  00001188   345          MVC   XCPM5(2),PRT3+14       fill in message with m4 field
                                                   346
0000043A  50F0 8288                     00000488   347          ST    R15,XCR15              save r15
0000043E  4100 004E                     0000004E   348          LA    R0,XCPLNG              message length
00000442  4110 8F16                     00001116   349          LA    R1,XCPLINE             message address
00000446  45F0 8402                     00000602   350          BAL   R15,RPTERROR
0000044A  58F0 8288                     00000488   351          L     R15,XCR15
                                                   352
0000044E  5800 856C                     0000076C   353          L     R0,=F'1'               set failed test indicator
00000452  5000 8E00                     00001000   354          ST    R0,FAILED
00000456  07FF                                     355          BR    R15                    return from xcheck
                                                   356
00000458                                           357          DS    0FD
00000458  00000000 00000000                        358 XCRESULT DS    XL16
00000468  00000000 00000000                        359 XCV1     DS    XL16
00000478  00000000 00000000                        360 XCV2     DS    XL16
00000488  00000000 00000000                        361 XCR15    DS    FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   363 ********************************************************************
                                                   364 * FPC not as expected:
                                                   365 *        issue message with test number, instruction under test,
                                                   366 *              expected FPC and received FPC
                                                   367 ********************************************************************
                                00000490  00000001 368 FPCMSG   EQU   *
00000490  5800 856C                       0000076C 369          L     R0,=F'1'                 set failed test indicator
00000494  5000 8E00                       00001000 370          ST    R0,FAILED
                                                   371
00000498  4820 5004                       00000004 372          LH    R2,TNUM                  get test number and convert
0000049C  4E20 8F90                       00001190 373          CVD   R2,DECNUM
000004A0  D211 8F7A 8F64        0000117A  00001164 374          MVC   PRT3,EDIT
000004A6  DE11 8F7A 8F90        0000117A  00001190 375          ED    PRT3,DECNUM
000004AC  D202 8E15 8F87        00001015  00001187 376          MVC   FPCPNUM(3),PRT3+13       fill in message with test #
                                                   377
000004B2  D207 8E39 5010        00001039  00000010 378          MVC   FPCPNAME,OPNAME          fill in message with instruction
                                                   379 *
000004B8  B982 0022                                380          XGR   R2,R2
000004BC  4320 500A                       0000000A 381          IC    R2,FLG                   get expected flags and convert
000004C0  4E20 8F90                       00001190 382          CVD   R2,DECNUM
000004C4  D211 8F7A 8F64        0000117A  00001164 383          MVC   PRT3,EDIT
000004CA  DE11 8F7A 8F90        0000117A  00001190 384          ED    PRT3,DECNUM
000004D0  D202 8E52 8F87        00001052  00001187 385          MVC   FPCPEXP(3),PRT3+13       fill in message with m3 field
                                                   386 *
000004D6  B982 0022                                387          XGR   R2,R2
000004DA  4320 5041                       00000041 388          IC    R2,FPC_R+1               get received flags and convert
000004DE  4E20 8F90                       00001190 389          CVD   R2,DECNUM
000004E2  D211 8F7A 8F64        0000117A  00001164 390          MVC   PRT3,EDIT
000004E8  DE11 8F7A 8F90        0000117A  00001190 391          ED    PRT3,DECNUM
000004EE  D202 8E67 8F87        00001067  00001187 392          MVC   FPCPGOT(3),PRT3+13       fill in message with FPC got
                                                   393 *
000004F4  4100 0063                       00000063 394          LA    R0,FPCPLNG               message length
000004F8  4110 8E08                       00001008 395          LA    R1,FPCPLINE              message address
000004FC  45F0 8402                       00000602 396          BAL   R15,RPTERROR
00000500  07F7                                     397          BR    R7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   399 ***********************************************************************
                                                   400 * VXC not as expected:
                                                   401 *        issue message with test number, instruction under test,
                                                   402 *              expected FPC and received FPC
                                                   403 ***********************************************************************
                                00000502  00000001 404 VXCMSG   EQU   *
00000502  5800 856C                       0000076C 405          L     R0,=F'1'              set failed test indicator
00000506  5000 8E00                       00001000 406          ST    R0,FAILED
                                                   407
0000050A  4820 5004                       00000004 408          LH    R2,TNUM               get test number and convert
0000050E  4E20 8F90                       00001190 409          CVD   R2,DECNUM
00000512  D211 8F7A 8F64      0000117A    00001164 410          MVC   PRT3,EDIT
00000518  DE11 8F7A 8F90      0000117A    00001190 411          ED    PRT3,DECNUM
0000051E  D202 8E78 8F87      00001078    00001187 412          MVC   VXCPNUM(3),PRT3+13    fill in message with test #
                                                   413
00000524  D207 8E9C 5010      0000109C    00000010 414          MVC   VXCPNAME,OPNAME       fill in message with instruction
                                                   415 *
0000052A  B982 0022                                416          XGR   R2,R2
0000052E  4320 500B                       0000000B 417          IC    R2,VXC                get expected flags and convert
00000532  4E20 8F90                       00001190 418          CVD   R2,DECNUM
00000536  D211 8F7A 8F64      0000117A    00001164 419          MVC   PRT3,EDIT
0000053C  DE11 8F7A 8F90      0000117A    00001190 420          ED    PRT3,DECNUM
00000542  D202 8EB5 8F87      000010B5    00001187 421          MVC   VXCPEXP(3),PRT3+13    fill in message with m3 field
                                                   422 *
00000548  B982 0022                                423          XGR   R2,R2
0000054C  4320 5042                       00000042 424          IC    R2,FPC_R+2            get received flags and convert
00000550  4E20 8F90                       00001190 425          CVD   R2,DECNUM
00000554  D211 8F7A 8F64      0000117A    00001164 426          MVC   PRT3,EDIT
0000055A  DE11 8F7A 8F90      0000117A    00001190 427          ED    PRT3,DECNUM
00000560  D202 8ECA 8F87      000010CA    00001187 428          MVC   VXCPGOT(3),PRT3+13    fill in message with FPC got
                                                   429 *
00000566  4100 0063                       00000063 430          LA    R0,VXCPLNG            message length
0000056A  4110 8E6B                       0000106B 431          LA    R1,VXCPLINE           message address
0000056E  45F0 8402                       00000602 432          BAL   R15,RPTERROR
00000572  07F7                                     433          BR    R7
                                                   434
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   436 ***********************************************************************
                                                   437 * result not as expected:
                                                   438 *        issue message with test number, instruction under test
                                                   439 *              and instruction m4
                                                   440 ***********************************************************************
                                00000574  00000001 441 FAILMSG  EQU   *
00000574  5800 856C                       0000076C 442          L     R0,=F'1'                set failed test indicator
00000578  5000 8E00                       00001000 443          ST    R0,FAILED
                                                   444
0000057C  4820 5004                       00000004 445          LH    R2,TNUM                 get test number and convert
00000580  4E20 8F90                       00001190 446          CVD   R2,DECNUM
00000584  D211 8F7A 8F64        0000117A  00001164 447          MVC   PRT3,EDIT
0000058A  DE11 8F7A 8F90        0000117A  00001190 448          ED    PRT3,DECNUM
00000590  D202 8EDB 8F87        000010DB  00001187 449          MVC   PRTNUM(3),PRT3+13       fill in message with test #
                                                   450
00000596  D207 8EF6 5010        000010F6  00000010 451          MVC   PRTNAME,OPNAME          fill in message with instruction
                                                   452 *
0000059C  B982 0022                                453          XGR   R2,R2
000005A0  4320 5008                       00000008 454          IC    R2,M4                   get m3 and convert
000005A4  4E20 8F90                       00001190 455          CVD   R2,DECNUM
000005A8  D211 8F7A 8F64        0000117A  00001164 456          MVC   PRT3,EDIT
000005AE  DE11 8F7A 8F90        0000117A  00001190 457          ED    PRT3,DECNUM
000005B4  D201 8F07 8F88        00001107  00001188 458          MVC   PRTM4(2),PRT3+14        fill in message with m3 field
                                                   459 *
000005BA  B982 0022                                460          XGR   R2,R2
000005BE  4320 5009                       00000009 461          IC    R2,M5                   get m4 and convert
000005C2  4E20 8F90                       00001190 462          CVD   R2,DECNUM
000005C6  D211 8F7A 8F64        0000117A  00001164 463          MVC   PRT3,EDIT
000005CC  DE11 8F7A 8F90        0000117A  00001190 464          ED    PRT3,DECNUM
000005D2  D201 8F13 8F88        00001113  00001188 465          MVC   PRTM5(2),PRT3+14        fill in message with m4 field
                                                   466 *
000005D8  4100 0048                       00000048 467          LA    R0,PRTLNG               message length
000005DC  4110 8ECE                       000010CE 468          LA    R1,PRTLINE              message address
000005E0  45F0 8402                       00000602 469          BAL   R15,RPTERROR

                                                   471 ***********************************************************************
                                                   472 * continue after a failed test
                                                   473 ***********************************************************************
                                000005E4  00000001 474 FAILCONT EQU   *
000005E4  5800 856C                       0000076C 475          L     R0,=F'1'                set failed test indicator
000005E8  5000 8E00                       00001000 476          ST    R0,FAILED
                                                   477
000005EC  41C0 C004                       00000004 478          LA    R12,4(0,R12)            next test address
000005F0  47F0 80EC                       000002EC 479          B     NEXTE6

                                                   481 ***********************************************************************
                                                   482 * end of testing; set ending psw
                                                   483 ***********************************************************************
                                000005F4  00000001 484 ENDTEST  EQU   *
000005F4  5810 8E00                       00001000 485          L     R1,FAILED               did a test fail?
000005F8  1211                                     486          LTR   R1,R1
000005FA  4780 8508                       00000708 487          BZ    EOJ                      No, exit
000005FE  47F0 8520                       00000720 488          B     FAILTEST                 Yes, exit with BAD PSW
                                                   489
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   491 ***********************************************************************
                                                   492 *        RPTERROR          Report instruction test in error
                                                   493 *                             R0 = MESSGAE LENGTH
                                                   494 *                             R1 = ADDRESS OF MESSAGE
                                                   495 ***********************************************************************

00000602  50F0 8420                    00000620    497 RPTERROR ST    R15,RPTSAVE             Save return address
00000606  5050 8424                    00000624    498          ST    R5,RPTSVR5              Save R5
                                                   499 *
                                                   500 *        Use Hercules Diagnose for Message to console
                                                   501 *
0000060A  9002 8428                    00000628    502          STM   R0,R2,RPTDWSAV          save regs used by MSG
0000060E  4520 8438                    00000638    503          BAL   R2,MSG                  call Hercules console MSG display
00000612  9802 8428                    00000628    504          LM    R0,R2,RPTDWSAV          restore regs


00000616  5850 8424                    00000624    506          L     R5,RPTSVR5              Restore R5
0000061A  58F0 8420                    00000620    507          L     R15,RPTSAVE             Restore return address
0000061E  07FF                                     508          BR    R15                     Return to caller

00000620  00000000                                 510 RPTSAVE  DC    F'0'                    R15 save area
00000624  00000000                                 511 RPTSVR5  DC    F'0'                    R5 save area

00000628  00000000 00000000                        513 RPTDWSAV DC    2D'0'                   R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   515 ***********************************************************************
                                                   516 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                   517 *              R2 = return address
                                                   518 ***********************************************************************

00000638  4900 8570                      00000770  520 MSG      CH    R0,=H'0'                Do we even HAVE a message?
0000063C  07D2                                     521          BNHR  R2                      No, ignore

0000063E  9002 8474                      00000674  523          STM   R0,R2,MSGSAVE           Save registers

00000642  4900 8572                      00000772  525          CH    R0,=AL2(L'MSGMSG)       Message length within limits?
00000646  47D0 844E                      0000064E  526          BNH   MSGOK                   Yes, continue
0000064A  4100 0064                      00000064  527          LA    R0,L'MSGMSG             No, set to maximum

0000064E  1820                                     529 MSGOK    LR    R2,R0                   Copy length to work register
00000650  0620                                     530          BCTR  R2,0                    Minus-1 for execute
00000652  4420 8480                      00000680  531          EX    R2,MSGMVC               Copy message to O/P buffer

00000656  4120 200A                      0000000A  533          LA    R2,1+L'MSGCMD(,R2)      Calculate true command length
0000065A  4110 8486                      00000686  534          LA    R1,MSGCMD               Point to true command

0000065E  83120008                                 536          DC    X'83',X'12',X'0008'     Issue Hercules Diagnose X'008'
00000662  4780 846E                      0000066E  537          BZ    MSGRET                  Return if successful
                                                   538
00000666  1222                                     539          LTR   R2,R2                   Is Diag8 Ry (R2) 0?
00000668  4780 846E                      0000066E  540          BZ    MSGRET                    an error occurred but coninue
                                                   541
0000066C  0000                                     542          DC    H'0'                    CRASH for debugging purposes

0000066E  9802 8474                      00000674  544 MSGRET   LM    R0,R2,MSGSAVE           Restore registers
00000672  07F2                                     545          BR    R2                      Return to caller




00000674  00000000 00000000                        547 MSGSAVE  DC    3F'0'                   Registers save area
00000680  D200 848F 1000       0000068F  00000000  548 MSGMVC   MVC   MSGMSG(0),0(R1)         Executed instruction


00000686  D4E2C7D5 D6C8405C                        550 MSGCMD   DC    C'MSGNOH * '            *** HERCULES MESSAGE COMMAND ***
0000068F  40404040 40404040                        551 MSGMSG   DC    CL100' '                 The message text to be displayed
                                                   552
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   554 ***********************************************************************
                                                   555 *        Normal completion or Abnormal termination PSWs
                                                   556 ***********************************************************************

000006F8  00020001 80000000                        558 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

00000708  B2B2 84F8                  000006F8      560 EOJ      LPSWE EOJPSW                  Normal completion

00000710  00020001 80000000                        562 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

00000720  B2B2 8510                  00000710      564 FAILTEST LPSWE FAILPSW                 Abnormal termination

                                                   566 ***********************************************************************
                                                   567 *        Working Storage
                                                   568 ***********************************************************************

00000724  00000000                                 570 CTLR0    DS    F                       CR0
00000728  00000000                                 571          DS    F
0000072C  00000000                                 572 FPCMOFF  DC    XL4'00000000'           FPC masks off
00000730  F8000000                                 573 FPCMON   DC    XL4'F8000000'           FPC masks on
00000734  80000000                                 574 FPCMIO   DC    XL4'80000000'           FPC mask invalid operation
00000738  20000000                                 575 FPCMOV   DC    XL4'20000000'           FPC mask overflow
0000073C  10000000                                 576 FPCMUN   DC    XL4'10000000'           FPC mask underflow
00000740  08000000                                 577 FPCMIE   DC    XL4'08000000'           FPC mask inexact

                                                   579
00000748                                           580          LTORG ,                 Literals pool
00000748  E2D2C9D7 40E7C340                        581                =CL8'SKIP XC '
00000750  40404040 40404040                        582                =CL8'        '
00000758  E5C3D9D5 C6404040                        583                =CL8'VCRNF'
00000760  00000004                                 584                =F'4'
00000764  00002D04                                 585                =A(E6TESTS)
00000768  00000008                                 586                =XL4'00000008'
0000076C  00000001                                 587                =F'1'
00000770  0000                                     588                =H'0'
00000772  0064                                     589                =AL2(L'MSGMSG)
00000774  E2                                       590                =CL1'S'
                                                   591
                                                   592 *        some constants
                                                   593
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                              00000400  00000001   594 K        EQU   1024             One KB
                              00001000  00000001   595 PAGE     EQU   (4*K)            Size of one page
                              00010000  00000001   596 K64      EQU   (64*K)           64 KB
                              00100000  00000001   597 MB       EQU   (K*K)             1 MB
                                                   598
                                                   599
                              AABBCCDD  00000001   600 REG2PATT EQU   X'AABBCCDD'    Polluted Register pattern
                              000000DD  00000001   601 REG2LOW  EQU         X'DD'    (last byte above)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   603 *=======================================================================
                                                   604 *
                                                   605 *  NOTE: start data on an address that is easy to display
                                                   606 *        within Hercules
                                                   607 *
                                                   608 *=======================================================================
                                                   609
00000775                     00000775  00001000    610          ORG   ZVE6TST+X'1000'
00001000  00000000                                 611 FAILED   DC    F'0'                    some test failed?
00001004  00000000                                 612 TESTING  DC    F'0'                    current test #

                                                   614
                                                   615 ***********************************************************************
                                                   616 *        TEST failed : FPC message
                                                   617 ***********************************************************************
                                                   618 *
                                                   619 *        failed message and associated editting
                                                   620 *
00001008  40404040 4040E385                        621 FPCPLINE DC    C'      Test # '
00001015  A7A7A7                                   622 FPCPNUM  DC    C'xxx'
00001018  40A69996 958740C6                        623          DC    c' wrong FPC flags for instruction '
00001039  A7A7A7A7 A7A7A7A7                        624 FPCPNAME DC    CL8'xxxxxxxx'
00001041  4085A797 8583A385                        625          DC    C' expected: flags='
00001052  A7A7A7                                   626 FPCPEXP  DC    C'xxx'
00001055  6B                                       627          DC    C','
00001056  40998583 8589A585                        628          DC    C' received: flags='
00001067  A7A7A7                                   629 FPCPGOT  DC    C'xxx'
0000106A  4B                                       630          DC    C'.'
                             00000063  00000001    631 FPCPLNG  EQU   *-FPCPLINE

                                                   633
                                                   634 ***********************************************************************
                                                   635 *        TEST failed : VXC message
                                                   636 ***********************************************************************
                                                   637 *
                                                   638 *        failed message and associated editting
                                                   639 *
0000106B  40404040 4040E385                        640 VXCPLINE DC    C'      Test # '
00001078  A7A7A7                                   641 VXCPNUM  DC    C'xxx'
0000107B  40A69996 958740C6                        642          DC    c' wrong FPC VXC   for instruction '
0000109C  A7A7A7A7 A7A7A7A7                        643 VXCPNAME DC    CL8'xxxxxxxx'
000010A4  4085A797 8583A385                        644          DC    C' expected:   VXC='
000010B5  A7A7A7                                   645 VXCPEXP  DC    C'xxx'
000010B8  6B                                       646          DC    C','
000010B9  40998583 8589A585                        647          DC    C' received:   VXC='
000010CA  A7A7A7                                   648 VXCPGOT  DC    C'xxx'
000010CD  4B                                       649          DC    C'.'
                             00000063  00000001    650 VXCPLNG  EQU   *-VXCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   652
                                                   653 ********************************************************************
                                                   654 *        TEST failed : result messgae
                                                   655 ********************************************************************
                                                   656 *
                                                   657 *        failed message and associated editting
                                                   658 *
000010CE  40404040 4040E385                        659 PRTLINE  DC    C'       Test # '
000010DB  A7A7A7                                   660 PRTNUM   DC    C'xxx'
000010DE  40868189 93858440                        661          DC    c' failed for instruction '
000010F6  A7A7A7A7 A7A7A7A7                        662 PRTNAME  DC    CL8'xxxxxxxx'
000010FE  40A689A3 884094F4                        663          DC    C' with m4='
00001107  A7A7                                     664 PRTM4    DC    C'xx'
00001109  6B                                       665          DC    C','
0000110A  40A689A3 884094F5                        666          DC    C' with m5='
00001113  A7A7                                     667 PRTM5    DC    C'xx'
00001115  4B                                       668          DC    C'.'
                             00000048  00000001    669 PRTLNG   EQU   *-PRTLINE

                                                   671 ********************************************************************
                                                   672 *        TEST failed : XCHECK
                                                   673 ********************************************************************
                                                   674 *
                                                   675 *        XCHECK failed message
                                                   676 *
00001116  40404040 4040E385                        677 XCPLINE  DC    C'       Test # '
00001123  A7A7A7                                   678 XCPTNUM  DC    C'xxx'
00001126  40E7C3C8 C5C3D240                        679          DC    c' XCHECK failed for instruction '
00001145  A7A7A7A7 A7A7A7A7                        680 XCPNAME  DC    CL8'xxxxxxxx'
0000114D  40A689A3 884094F4                        681          DC    C' with m4='
00001156  A7A7                                     682 XCPM4    DC    C'xx'
00001158  40A689A3 884094F5                        683          DC    C' with m5='
00001161  A7A7                                     684 XCPM5    DC    C'xx'
00001163  4B                                       685          DC    C'.'
                             0000004E  00000001    686 XCPLNG   EQU   *-XCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   688 **********************************************************************
                                                   689 *        TEST failed : message working storge
                                                   690 **********************************************************************
00001164  40212020 20202020                        691 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                   692
00001176  7E7E7E6E                                 693          DC    C'===>'
0000117A  40404040 40404040                        694 PRT3     DC    CL18' '
0000118C  4C7E7E7E                                 695          DC    C'<==='
00001190  00000000 00000000                        696 DECNUM   DS    CL16


                                                   698 *
                                                   699 *        Vector instruction results, pollution and input
                                                   700 *
000011A0  00000000 00000000                        701          DS    XL16                                      gap
000011B0  FFFFFFFF FFFFFFFF                        702 V1FUDGE  DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'    V1 FUDGE
000011C0  12345678 90123456                        703 V1INPUT  DC    XL16'1234567890123456789012345678901D'    V1 input
000011D0  00000000 00000000                        704          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   706 ***********************************************************************
                                                   707 *        E6TEST DSECT
                                                   708 ***********************************************************************

                                                   710 E6TEST   DSECT ,
00000000  00000000                                 711 TSUB     DC    A(0)               pointer  to test
00000004  0000                                     712 TNUM     DC    H'00'              Test Number
00000006  00                                       713          DC    X'00'
00000007  40                                       714 XCSKIP   DC    CL1' '             Y = skip cross check
00000008  00                                       715 M4       DC    HL1'00'            m4 used
00000009  00                                       716 M5       DC    HL1'00'            m5 used
0000000A  00                                       717 FLG      DC    X'00'              expected FPC flags
0000000B  00                                       718 VXC      DC    X'00'              VXC expected
0000000C  00000000                                 719 V2ADDR   DC    A(0)               address of v2: 16-byte packed decimal
00000010  40404040 40404040                        720 OPNAME   DC    CL8' '             E6 name
00000018  00000000                                 721 RELEN    DC    A(0)               result length
0000001C  00000000                                 722 READDR   DC    A(0)               expected result address
00000020  00000000 00000000                        723          DS    FD                    gap
00000028  00000000 00000000                        724 V1OUTPUT DS    XL16               V1 Output
00000038  00000000 00000000                        725          DS    FD                    gap
00000040  00000000                                 726 FPC_R    DS    F                  FPC after instruction
00000048  00000000 00000000                        727          DS    FD                    gap
00000050  40404040 40404040                        728 SKIPXC   DC    CL8' '             was cross check skipped?
00000058  00000000 00000000                        729          DS    FD                    gap
00000060  00000000 00000000                        730 XCOUTPUT DS    XL16               Cross check Output
00000070  00000000 00000000                        731          DS    XL16
00000080  00000000 00000000                        732          DS    FD                    gap
00000088  00000000                                 733 FPC_XC1  DS    F                  1st cross check FPC
0000008C  00000000                                 734 FPC_XC2  DS    F                  2nd cross check FPC
00000090  00000000 00000000                        735          DS    FD                    gap
00000098  00000000                                 736 WK1      DS    F                  debug area
0000009C  00000000                                 737 WK2      DS    F
000000A0                                           738          DS    0F
                                                   739 **
                                                   740 *        test routine will be here (from VRR-c macro)

                                 00000000  00002D83  742 ZVE6TST  CSECT ,
000011E0                                           743          DS    0F

                                                   745 ***********************************************************************
                                                   746 *     Macros to help build test tables
                                                   747 ***********************************************************************

                                                   749 *
                                                   750 * macro to generate individual test
                                                   751 *
                                                   752          MACRO
                                                   753          VRR_C &INST,&M4,&M5,&FLAGS,&VXC,&SKIP
                                                   754 .*                                  &INST  - VRR-c instruction under test
                                                   755 .*                                  &m4    - m4 field
                                                   756 .*                                  &m5    - m5 field
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   757 .*                                &flags - expected FPC flags (HEX)
                                                   758 .*                                &VXC   - expected VXC (HEX)
                                                   759 .*                                &SKIP  - S = skip cross check
                                                   760          GBLA  &TNUM
                                                   761 &TNUM    SETA  &TNUM+1
                                                   762
                                                   763          DS    0FD
                                                   764          USING *,R5               base for test data and test routine
                                                   765
                                                   766 T&TNUM   DC    A(X&TNUM)          address of test routine
                                                   767          DC    H'&TNUM'           test number
                                                   768          DC    X'00'
                                                   769          DC    CL1'&SKIP'         Y = skip cross check
                                                   770          DC    HL1'&M4'           m4
                                                   771          DC    HL1'&M5'           m5
                                                   772 FLG&TNUM DC    X'&FLAGS'          expected FPC flags
                                                   773 VXC&TNUM DC    X'&VXC'            expected VXC
                                                   774 V2_&TNUM DC    A(RE&TNUM+16)      address of v2: 16-byte packed decimal
                                                   775          DC    CL8'&INST'         instruction name
                                                   776          DC    A(16)              result length
                                                   777          DC    A(RE&TNUM)         address of expected resul
                                                   778          DS    FD                 gap
                                                   779 V1O&TNUM DS    XL16               V1 output
                                                   780          DS    FD                    gap
                                                   781 FPC_R_&TNUM DS F                  FPC after instruction
                                                   782          DS    FD                    gap
                                                   783          DC    CL8' '             was cross check skipped?
                                                   784          DS    FD                    gap
                                                   785 XCO&TNUM DS    XL16               Cross check Output
                                                   786          DS    XL16
                                                   787          DS    FD                    gap
                                                   788 FPC_XC_&TNUM DS F                1st cross check FPC
                                                   789          DS    F                 2nd cross check FPC
                                                   790          DS    FD                    gap
                                                   791          DS    F                  debug area
                                                   792          DS    F
                                                   793 .*
                                                   794 *
                                                   795 X&TNUM   DS    0F
                                                   796          LFPC  FPCMOFF            initialize FPC
                                                   797
                                                   798          LGF   R2,V2_&TNUM        get v2
                                                   799          VLM   V22,V23,0(R2)
                                                   800
                                                   801          &INST V22,V22,V23,&M4,&M5   test instruction (dest is source)
                                                   802
                                                   803          STFPC FPC_R_&TNUM        save FPC
                                                   804          VST   V22,V1O&TNUM       save instruction result
                                                   805
                                                   806          BR    R11                return
                                                   807
                                                   808 RE&TNUM  DS    0F                 expected 16 byte result
                                                   809          DROP  R5
                                                   810
                                                   811          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   813 *
                                                   814 * macro to generate table of pointers to individual tests
                                                   815 *
                                                   816          MACRO
                                                   817          PTTABLE
                                                   818          GBLA  &TNUM
                                                   819          LCLA  &CUR
                                                   820 &CUR     SETA  1
                                                   821 .*
                                                   822 TTABLE   DS    0F
                                                   823 .LOOP    ANOP
                                                   824 .*
                                                   825          DC    A(T&CUR)          TEST &CUR
                                                   826 .*
                                                   827 &CUR     SETA  &CUR+1
                                                   828          AIF   (&CUR LE &TNUM).LOOP
                                                   829 *
                                                   830          DC    A(0)              END OF TABLE
                                                   831          DC    A(0)
                                                   832 .*
                                                   833          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   835 ***********************************************************************
                                                   836 *        E6 VRR-c tests
                                                   837 ***********************************************************************
                                                   838          PRINT DATA
                                                   839 *
                                                   840 *        E675  VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                   841 *
                                                   842 *----------------------------------------------------------------------
                                                   843 *   VRR-c   instruction, m4, m5, flags, VXC, SKIP
                                                   844 *           NOTE: flags and VXC are HEX values
                                                   845 *              and VXC should always by 00 as the FPC mask is zero
                                                   846 *   followed by
                                                   847 *              v1 - 16 byte expected result
                                                   848 *              v2 - 16 byte source
                                                   849 *              v3 - 16 byte source
                                                   850 *----------------------------------------------------------------------
                                                   851 * VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                   852 *----------------------------------------------------------------------
                                                   853 * Short Float -> dlfloat  (with cross check dlfloat -> short float)
                                                   854 *
                                                   855 * some of these tests use numbers from PoP SA22-7832-13,
                                                   856 * Figure 9-2. Examples of Floating-Point Numbers ( page 9-6 )
                                                   857 *
                                                   858 * +0     simple instruction test and 'test' test
                                                   859          VRR_C VCRNF,0,2,00,00,N
000011E0                                           860+         DS    0FD
000011E0                     000011E0              861+         USING *,R5                base for test data and test routine
000011E0  00001280                                 862+T1       DC    A(X1)               address of test routine
000011E4  0001                                     863+         DC    H'1'                test number
000011E6  00                                       864+         DC    X'00'
000011E7  D5                                       865+         DC    CL1'N'              Y = skip cross check
000011E8  00                                       866+         DC    HL1'0'              m4
000011E9  02                                       867+         DC    HL1'2'              m5
000011EA  00                                       868+FLG1     DC    X'00'               expected FPC flags
000011EB  00                                       869+VXC1     DC    X'00'               expected VXC
000011EC  000012B4                                 870+V2_1     DC    A(RE1+16)           address of v2: 16-byte packed decimal
000011F0  E5C3D9D5 C6404040                        871+         DC    CL8'VCRNF'          instruction name
000011F8  00000010                                 872+         DC    A(16)               result length
000011FC  000012A4                                 873+         DC    A(RE1)              address of expected resul
00001200  00000000 00000000                        874+         DS    FD                  gap
00001208  00000000 00000000                        875+V1O1     DS    XL16                V1 output
00001210  00000000 00000000
00001218  00000000 00000000                        876+         DS    FD                     gap
00001220  00000000                                 877+FPC_R_1  DS    F                   FPC after instruction
00001228  00000000 00000000                        878+         DS    FD                     gap
00001230  40404040 40404040                        879+         DC    CL8' '              was cross check skipped?
00001238  00000000 00000000                        880+         DS    FD                     gap
00001240  00000000 00000000                        881+XCO1     DS    XL16                Cross check Output
00001248  00000000 00000000
00001250  00000000 00000000                        882+         DS    XL16
00001258  00000000 00000000
00001260  00000000 00000000                        883+         DS    FD                     gap
00001268  00000000                                 884+FPC_XC_1 DS    F                  1st cross check FPC
0000126C  00000000                                 885+         DS    F                  2nd cross check FPC
00001270  00000000 00000000                        886+         DS    FD                     gap
00001278  00000000                                 887+         DS    F                   debug area
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000127C  00000000                                 888+         DS    F
                                                   889+*
00001280                                           890+X1       DS    0F
00001280  B29D 852C                      0000072C  891+         LFPC  FPCMOFF            initialize FPC
00001284  E320 500C 0014                 000011EC  892+         LGF   R2,V2_1            get v2
0000128A  E767 2000 0C36                 00000000  893+         VLM   V22,V23,0(R2)
00001290  E666 7002 0E75                           894+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00001296  B29C 9020                      00001220  895+         STFPC FPC_R_1            save FPC
0000129A  E760 9008 080E                 00001208  896+         VST   V22,V1O1           save instruction result
000012A0  07FB                                     897+         BR    R11                return
000012A4                                           898+RE1      DS    0F                 expected 16 byte result
000012A4                                           899+         DROP  R5
000012A4  00000000 00000000                        900          DC    XL16'0000000000000000 0000000000000000'
000012AC  00000000 00000000
000012B4  00000000 00000000                        901          DC    XL16'0000000000000000 0000000000000000'
000012BC  00000000 00000000
000012C4  00000000 00000000                        902          DC    XL16'0000000000000000 0000000000000000'
000012CC  00000000 00000000
                                                   903
                                                   904 * +1, -1
                                                   905          VRR_C VCRNF,0,2,00,00,N
000012D8                                           906+         DS    0FD
000012D8                       000012D8            907+         USING *,R5               base for test data and test routine
000012D8  00001378                                 908+T2       DC    A(X2)              address of test routine
000012DC  0002                                     909+         DC    H'2'               test number
000012DE  00                                       910+         DC    X'00'
000012DF  D5                                       911+         DC    CL1'N'             Y = skip cross check
000012E0  00                                       912+         DC    HL1'0'             m4
000012E1  02                                       913+         DC    HL1'2'             m5
000012E2  00                                       914+FLG2     DC    X'00'              expected FPC flags
000012E3  00                                       915+VXC2     DC    X'00'              expected VXC
000012E4  000013AC                                 916+V2_2     DC    A(RE2+16)          address of v2: 16-byte packed decimal
000012E8  E5C3D9D5 C6404040                        917+         DC    CL8'VCRNF'         instruction name
000012F0  00000010                                 918+         DC    A(16)              result length
000012F4  0000139C                                 919+         DC    A(RE2)             address of expected resul
000012F8  00000000 00000000                        920+         DS    FD                 gap
00001300  00000000 00000000                        921+V1O2     DS    XL16               V1 output
00001308  00000000 00000000
00001310  00000000 00000000                        922+         DS    FD                    gap
00001318  00000000                                 923+FPC_R_2  DS    F                  FPC after instruction
00001320  00000000 00000000                        924+         DS    FD                    gap
00001328  40404040 40404040                        925+         DC    CL8' '             was cross check skipped?
00001330  00000000 00000000                        926+         DS    FD                    gap
00001338  00000000 00000000                        927+XCO2     DS    XL16               Cross check Output
00001340  00000000 00000000
00001348  00000000 00000000                        928+         DS    XL16
00001350  00000000 00000000
00001358  00000000 00000000                        929+         DS    FD                    gap
00001360  00000000                                 930+FPC_XC_2 DS    F                 1st cross check FPC
00001364  00000000                                 931+         DS    F                 2nd cross check FPC
00001368  00000000 00000000                        932+         DS    FD                    gap
00001370  00000000                                 933+         DS    F                  debug area
00001374  00000000                                 934+         DS    F
                                                   935+*
00001378                                           936+X2       DS    0F
00001378  B29D 852C                      0000072C  937+         LFPC  FPCMOFF            initialize FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000137C  E320 500C 0014                000012E4   938+         LGF   R2,V2_2             get v2
00001382  E767 2000 0C36                00000000   939+         VLM   V22,V23,0(R2)
00001388  E666 7002 0E75                           940+         VCRNF V22,V22,V23,0,2         test instruction (dest is source)
0000138E  B29C 5040                     00001318   941+         STFPC FPC_R_2             save FPC
00001392  E760 5028 080E                00001300   942+         VST   V22,V1O2            save instruction result
00001398  07FB                                     943+         BR    R11                 return
0000139C                                           944+RE2      DS    0F                  expected 16 byte result
0000139C                                           945+         DROP  R5
0000139C  3E000000 00000000                        946          DC    XL16'3E00000000000000 BE00000000000000'
000013A4  BE000000 00000000
000013AC  3F800000 00000000                        947          DC    XL16'3F80000000000000 0000000000000000'
000013B4  00000000 00000000
000013BC  BF800000 00000000                        948          DC    XL16'BF80000000000000 0000000000000000'
000013C4  00000000 00000000
                                                   949
                                                   950 * +.5, -.5
                                                   951          VRR_C VCRNF,0,2,00,00,N
000013D0                                           952+         DS    0FD
000013D0                      000013D0             953+         USING *,R5                base for test data and test routine
000013D0  00001470                                 954+T3       DC    A(X3)               address of test routine
000013D4  0003                                     955+         DC    H'3'                test number
000013D6  00                                       956+         DC    X'00'
000013D7  D5                                       957+         DC    CL1'N'              Y = skip cross check
000013D8  00                                       958+         DC    HL1'0'              m4
000013D9  02                                       959+         DC    HL1'2'              m5
000013DA  00                                       960+FLG3     DC    X'00'               expected FPC flags
000013DB  00                                       961+VXC3     DC    X'00'               expected VXC
000013DC  000014A4                                 962+V2_3     DC    A(RE3+16)           address of v2: 16-byte packed decimal
000013E0  E5C3D9D5 C6404040                        963+         DC    CL8'VCRNF'          instruction name
000013E8  00000010                                 964+         DC    A(16)               result length
000013EC  00001494                                 965+         DC    A(RE3)              address of expected resul
000013F0  00000000 00000000                        966+         DS    FD                  gap
000013F8  00000000 00000000                        967+V1O3     DS    XL16                V1 output
00001400  00000000 00000000
00001408  00000000 00000000                        968+         DS    FD                     gap
00001410  00000000                                 969+FPC_R_3  DS    F                   FPC after instruction
00001418  00000000 00000000                        970+         DS    FD                     gap
00001420  40404040 40404040                        971+         DC    CL8' '              was cross check skipped?
00001428  00000000 00000000                        972+         DS    FD                     gap
00001430  00000000 00000000                        973+XCO3     DS    XL16                Cross check Output
00001438  00000000 00000000
00001440  00000000 00000000                        974+         DS    XL16
00001448  00000000 00000000
00001450  00000000 00000000                        975+         DS    FD                     gap
00001458  00000000                                 976+FPC_XC_3 DS    F                  1st cross check FPC
0000145C  00000000                                 977+         DS    F                  2nd cross check FPC
00001460  00000000 00000000                        978+         DS    FD                     gap
00001468  00000000                                 979+         DS    F                   debug area
0000146C  00000000                                 980+         DS    F
                                                   981+*
00001470                                           982+X3       DS    0F
00001470  B29D 852C                     0000072C   983+         LFPC  FPCMOFF             initialize FPC
00001474  E320 500C 0014                000013DC   984+         LGF   R2,V2_3             get v2
0000147A  E767 2000 0C36                00000000   985+         VLM   V22,V23,0(R2)
00001480  E666 7002 0E75                           986+         VCRNF V22,V22,V23,0,2         test instruction (dest is source)
00001486  B29C 5040                     00001410   987+         STFPC FPC_R_3             save FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000148A  E760 5028 080E                000013F8   988+         VST   V22,V103          save instruction result
00001490  07FB                                     989+         BR    R11               return
00001494                                           990+RE3      DS    0F                expected 16 byte result
00001494                                           991+         DROP  R5
00001494  3C000000 00000000                        992          DC    XL16'3C00000000000000 BC00000000000000'
0000149C  BC000000 00000000
000014A4  3F000000 00000000                        993          DC    XL16'3F00000000000000 0000000000000000'
000014AC  00000000 00000000
000014B4  BF000000 00000000                        994          DC    XL16'BF00000000000000 0000000000000000'
000014BC  00000000 00000000
                                                   995
                                                   996 * +1/64, -1/64
                                                   997          VRR_C VCRNF,0,2,00,00,N
000014C8                                           998+         DS    0FD
000014C8                      000014C8             999+         USING *,R5              base for test data and test routine
000014C8  00001568                                1000+T4       DC    A(X4)             address of test routine
000014CC  0004                                    1001+         DC    H'4'              test number
000014CE  00                                      1002+         DC    X'00'
000014CF  D5                                      1003+         DC    CL1'N'            Y = skip cross check
000014D0  00                                      1004+         DC    HL1'0'            m4
000014D1  02                                      1005+         DC    HL1'2'            m5
000014D2  00                                      1006+FLG4     DC    X'00'             expected FPC flags
000014D3  00                                      1007+VXC4     DC    X'00'             expected VXC
000014D4  0000159C                                1008+V2_4     DC    A(RE4+16)         address of v2: 16-byte packed decimal
000014D8  E5C3D9D5 C6404040                       1009+         DC    CL8'VCRNF'        instruction name
000014E0  00000010                                1010+         DC    A(16)             result length
000014E4  0000158C                                1011+         DC    A(RE4)            address of expected resul
000014E8  00000000 00000000                       1012+         DS    FD                gap
000014F0  00000000 00000000                       1013+V1O4     DS    XL16              V1 output
000014F8  00000000 00000000
00001500  00000000 00000000                       1014+         DS    FD                   gap
00001508  00000000                                1015+FPC_R_4  DS    F                 FPC after instruction
00001510  00000000 00000000                       1016+         DS    FD                   gap
00001518  40404040 40404040                       1017+         DC    CL8' '            was cross check skipped?
00001520  00000000 00000000                       1018+         DS    FD                   gap
00001528  00000000 00000000                       1019+XCO4     DS    XL16              Cross check Output
00001530  00000000 00000000
00001538  00000000 00000000                       1020+         DS    XL16
00001540  00000000 00000000
00001548  00000000 00000000                       1021+         DS    FD                   gap
00001550  00000000                                1022+FPC_XC_4 DS    F                1st cross check FPC
00001554  00000000                                1023+         DS    F                2nd cross check FPC
00001558  00000000 00000000                       1024+         DS    FD                   gap
00001560  00000000                                1025+         DS    F                 debug area
00001564  00000000                                1026+         DS    F
                                                  1027+*
00001568                                          1028+X4       DS    0F
00001568  B29D 852C                     0000072C  1029+         LFPC  FPCMOFF           initialize FPC
0000156C  E320 500C 0014                000014D4  1030+         LGF   R2,V2_4           get v2
00001572  E767 2000 0C36                00000000  1031+         VLM   V22,V23,0(R2)
00001578  E666 7002 0E75                          1032+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000157E  B29C 5040                     00001508  1033+         STFPC FPC_R_4           save FPC
00001582  E760 5028 080E                000014F0  1034+         VST   V22,V1O4          save instruction result
00001588  07FB                                    1035+         BR    R11               return
0000158C                                          1036+RE4      DS    0F                expected 16 byte result
0000158C                                          1037+         DROP  R5
```

```
  LOC        OBJECT CODE        ADDR1     ADDR2     STMT

0000158C  32000000 00000000                        1038          DC    XL16'3200000000000000 B200000000000000'
00001594  B2000000 00000000
0000159C  3C800000 00000000                        1039          DC    XL16'3C80000000000000 0000000000000000'
000015A4  00000000 00000000
000015AC  BC800000 00000000                        1040          DC    XL16'BC80000000000000 0000000000000000'
000015B4  00000000 00000000
                                                   1041
                                                   1042 * +0, -0
                                                   1043          VRR_C VCRNF,0,2,00,00,N
000015C0                                           1044+         DS    0FD
000015C0                      000015C0             1045+         USING *,R5              base for test data and test routine
000015C0  00001660                                 1046+T5       DC    A(X5)             address of test routine
000015C4  0005                                     1047+         DC    H'5'              test number
000015C6  00                                       1048+         DC    X'00'
000015C7  D5                                       1049+         DC    CL1'N'            Y = skip cross check
000015C8  00                                       1050+         DC    HL1'0'            m4
000015C9  02                                       1051+         DC    HL1'2'            m5
000015CA  00                                       1052+FLG5     DC    X'00'             expected FPC flags
000015CB  00                                       1053+VXC5     DC    X'00'             expected VXC
000015CC  00001694                                 1054+V2_5     DC    A(RE5+16)         address of v2: 16-byte packed decimal
000015D0  E5C3D9D5 C6404040                        1055+         DC    CL8'VCRNF'        instruction name
000015D8  00000010                                 1056+         DC    A(16)             result length
000015DC  00001684                                 1057+         DC    A(RE5)            address of expected resul
000015E0  00000000 00000000                        1058+         DS    FD                gap
000015E8  00000000 00000000                        1059+V1O5     DS    XL16              V1 output
000015F0  00000000 00000000
000015F8  00000000 00000000                        1060+         DS    FD                   gap
00001600  00000000                                 1061+FPC_R_5  DS    F                 FPC after instruction
00001608  00000000 00000000                        1062+         DS    FD                   gap
00001610  40404040 40404040                        1063+         DC    CL8' '            was cross check skipped?
00001618  00000000 00000000                        1064+         DS    FD                   gap
00001620  00000000 00000000                        1065+XCO5     DS    XL16              Cross check Output
00001628  00000000 00000000
00001630  00000000 00000000                        1066+         DS    XL16
00001638  00000000 00000000
00001640  00000000 00000000                        1067+         DS    FD                   gap
00001648  00000000                                 1068+FPC_XC_5 DS    F                1st cross check FPC
0000164C  00000000                                 1069+         DS    F                2nd cross check FPC
00001650  00000000 00000000                        1070+         DS    FD                   gap
00001658  00000000                                 1071+         DS    F                 debug area
0000165C  00000000                                 1072+         DS    F
                                                   1073+*
00001660                                           1074+X5       DS    0F
00001660  B29D 852C                     0000072C   1075+         LFPC  FPCMOFF           initialize FPC
00001664  E320 500C 0014                000015CC   1076+         LGF   R2,V2_5           get v2
0000166A  E767 2000 0C36                00000000   1077+         VLM   V22,V23,0(R2)
00001670  E666 7002 0E75                           1078+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001676  B29C 5040                     00001600   1079+         STFPC FPC_R_5           save FPC
0000167A  E760 5028 080E                000015E8   1080+         VST   V22,V1O5          save instruction result
00001680  07FB                                     1081+         BR    R11               return
00001684                                           1082+RE5      DS    0F                expected 16 byte result
00001684                                           1083+         DROP  R5
00001684  00000000 00000000                        1084          DC    XL16'0000000000000000 8000000000000000'
0000168C  80000000 00000000
00001694  00000000 00000000                        1085          DC    XL16'0000000000000000 0000000000000000'
0000169C  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000016A4  80000000 00000000                        1086          DC    XL16'8000000000000000 0000000000000000'
000016AC  00000000 00000000
                                                   1087
                                                   1088 * +15, -15
                                                   1089          VRR_C VCRNF,0,2,00,00,N
000016B8                                           1090+         DS    0FD
000016B8                     000016B8              1091+         USING *,R5                base for test data and test routine
000016B8  00001758                                 1092+T6       DC    A(X6)               address of test routine
000016BC  0006                                     1093+         DC    H'6'                test number
000016BE  00                                       1094+         DC    X'00'
000016BF  D5                                       1095+         DC    CL1'N'              Y = skip cross check
000016C0  00                                       1096+         DC    HL1'0'              m4
000016C1  02                                       1097+         DC    HL1'2'              m5
000016C2  00                                       1098+FLG6     DC    X'00'               expected FPC flags
000016C3  00                                       1099+VXC6     DC    X'00'               expected VXC
000016C4  0000178C                                 1100+V2_6     DC    A(RE6+16)           address of v2: 16-byte packed decimal
000016C8  E5C3D9D5 C6404040                        1101+         DC    CL8'VCRNF'          instruction name
000016D0  00000010                                 1102+         DC    A(16)               result length
000016D4  0000177C                                 1103+         DC    A(RE6)              address of expected resul
000016D8  00000000 00000000                        1104+         DS    FD                  gap
000016E0  00000000 00000000                        1105+V1O6     DS    XL16                V1 output
000016E8  00000000 00000000
000016F0  00000000 00000000                        1106+         DS    FD                     gap
000016F8  00000000                                 1107+FPC_R_6  DS    F                   FPC after instruction
00001700  00000000 00000000                        1108+         DS    FD                     gap
00001708  40404040 40404040                        1109+         DC    CL8' '              was cross check skipped?
00001710  00000000 00000000                        1110+         DS    FD                     gap
00001718  00000000 00000000                        1111+XCO6     DS    XL16                Cross check Output
00001720  00000000 00000000
00001728  00000000 00000000                        1112+         DS    XL16
00001730  00000000 00000000
00001738  00000000 00000000                        1113+         DS    FD                     gap
00001740  00000000                                 1114+FPC_XC_6 DS    F                  1st cross check FPC
00001744  00000000                                 1115+         DS    F                  2nd cross check FPC
00001748  00000000 00000000                        1116+         DS    FD                     gap
00001750  00000000                                 1117+         DS    F                   debug area
00001754  00000000                                 1118+         DS    F
                                                   1119+*
00001758                                           1120+X6       DS    0F
00001758  B29D 852C                    0000072C    1121+         LFPC  FPCMOFF             initialize FPC
0000175C  E320 500C 0014               000016C4    1122+         LGF   R2,V2_6             get v2
00001762  E767 2000 0C36               00000000    1123+         VLM   V22,V23,0(R2)
00001768  E666 7002 0E75                           1124+         VCRNF V22,V22,V23,0,2         test instruction (dest is source)
0000176E  B29C 5040                    000016F8    1125+         STFPC FPC_R_6             save FPC
00001772  E760 5028 080E               000016E0    1126+         VST   V22,V1O6            save instruction result
00001778  07FB                                     1127+         BR    R11                 return
0000177C                                           1128+RE6      DS    0F                  expected 16 byte result
0000177C                                           1129+         DROP  R5
0000177C  45C00000 00000000                        1130          DC    XL16'45C0000000000000 C5C0000000000000'
00001784  C5C00000 00000000
0000178C  41700000 00000000                        1131          DC    XL16'4170000000000000 0000000000000000'
00001794  00000000 00000000
0000179C  C1700000 00000000                        1132          DC    XL16'C170000000000000 0000000000000000'
000017A4  00000000 00000000
                                                   1133
                                                   1134 * +20/7, - 20/7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1135          VRR_C VCRNF,0,2,00,00,S                 skip xc: lost digits
000017B0                                           1136+         DS    0FD
000017B0                       000017B0            1137+         USING *,R5                base for test data and test routine
000017B0  00001850                                 1138+T7       DC    A(X7)               address of test routine
000017B4  0007                                     1139+         DC    H'7'                test number
000017B6  00                                       1140+         DC    X'00'
000017B7  E2                                       1141+         DC    CL1'S'              Y = skip cross check
000017B8  00                                       1142+         DC    HL1'0'              m4
000017B9  02                                       1143+         DC    HL1'2'              m5
000017BA  00                                       1144+FLG7     DC    X'00'               expected FPC flags
000017BB  00                                       1145+VXC7     DC    X'00'               expected VXC
000017BC  00001884                                 1146+V2_7     DC    A(RE7+16)           address of v2: 16-byte packed decimal
000017C0  E5C3D9D5 C6404040                        1147+         DC    CL8'VCRNF'          instruction name
000017C8  00000010                                 1148+         DC    A(16)               result length
000017CC  00001874                                 1149+         DC    A(RE7)              address of expected resul
000017D0  00000000 00000000                        1150+         DS    FD                  gap
000017D8  00000000 00000000                        1151+V1O7     DS    XL16                V1 output
000017E0  00000000 00000000
000017E8  00000000 00000000                        1152+         DS    FD                     gap
000017F0  00000000                                 1153+FPC_R_7  DS    F                   FPC after instruction
000017F8  00000000 00000000                        1154+         DS    FD                     gap
00001800  40404040 40404040                        1155+         DC    CL8' '              was cross check skipped?
00001808  00000000 00000000                        1156+         DS    FD                     gap
00001810  00000000 00000000                        1157+XCO7     DS    XL16                Cross check Output
00001818  00000000 00000000
00001820  00000000 00000000                        1158+         DS    XL16
00001828  00000000 00000000
00001830  00000000 00000000                        1159+         DS    FD                     gap
00001838  00000000                                 1160+FPC_XC_7 DS    F                  1st cross check FPC
0000183C  00000000                                 1161+         DS    F                  2nd cross check FPC
00001840  00000000 00000000                        1162+         DS    FD                     gap
00001848  00000000                                 1163+         DS    F                   debug area
0000184C  00000000                                 1164+         DS    F
                                                   1165+*
00001850                                           1166+X7       DS    0F
00001850  B29D 852C                      0000072C  1167+         LFPC  FPCMOFF             initialize FPC
00001854  E320 500C 0014                 000017BC  1168+         LGF   R2,V2_7             get v2
0000185A  E767 2000 0C36                 00000000  1169+         VLM   V22,V23,0(R2)
00001860  E666 7002 0E75                           1170+         VCRNF V22,V22,V23,0,2         test instruction (dest is source)
00001866  B29C 5040                      000017F0  1171+         STFPC FPC_R_7             save FPC
0000186A  E760 5028 080E                 000017D8  1172+         VST   V22,V1O7            save instruction result
00001870  07FB                                     1173+         BR    R11                 return
00001874                                           1174+RE7      DS    0F                  expected 16 byte result
00001874                                           1175+         DROP  R5
00001874  40DB0000 00000000                        1176          DC    XL16'40DB000000000000 C0DB000000000000'
0000187C  C0DB0000 00000000
00001884  4036DB6E 00000000                        1177          DC    XL16'4036DB6E00000000 0000000000000000'
0000188C  00000000 00000000
00001894  C036DB6E 00000000                        1178          DC    XL16'C036DB6E00000000 0000000000000000'
0000189C  00000000 00000000
                                                   1179
                                                   1180 * +2^(-126), -2^(-126)
                                                   1181          VRR_C VCRNF,0,2,10,00,S                 skip xc: underflow
000018A8                                           1182+         DS    0FD
000018A8                       000018A8            1183+         USING *,R5                base for test data and test routine
000018A8  00001948                                 1184+T8       DC    A(X8)               address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000018AC  0008                                    1185+          DC    H'8'                test number
000018AE  00                                      1186+          DC    X'00'
000018AF  E2                                      1187+          DC    CL1'S'              Y = skip cross check
000018B0  00                                      1188+          DC    HL1'0'              m4
000018B1  02                                      1189+          DC    HL1'2'              m5
000018B2  10                                      1190+FLG8      DC    X'10'               expected FPC flags
000018B3  00                                      1191+VXC8      DC    X'00'               expected VXC
000018B4  0000197C                                1192+V2_8      DC    A(RE8+16)           address of v2: 16-byte packed decimal
000018B8  E5C3D9D5 C6404040                       1193+          DC    CL8'VCRNF'          instruction name
000018C0  00000010                                1194+          DC    A(16)               result length
000018C4  0000196C                                1195+          DC    A(RE8)              address of expected resul
000018C8  00000000 00000000                       1196+          DS    FD                  gap
000018D0  00000000 00000000                       1197+V1O8      DS    XL16                V1 output
000018D8  00000000 00000000
000018E0  00000000 00000000                       1198+          DS    FD                     gap
000018E8  00000000                                1199+FPC_R_8   DS    F                   FPC after instruction
000018F0  00000000 00000000                       1200+          DS    FD                     gap
000018F8  40404040 40404040                       1201+          DC    CL8' '              was cross check skipped?
00001900  00000000 00000000                       1202+          DS    FD                     gap
00001908  00000000 00000000                       1203+XCO8      DS    XL16                Cross check Output
00001910  00000000 00000000
00001918  00000000 00000000                       1204+          DS    XL16
00001920  00000000 00000000
00001928  00000000 00000000                       1205+          DS    FD                     gap
00001930  00000000                                1206+FPC_XC_8 DS     F                  1st cross check FPC
00001934  00000000                                1207+          DS    F                  2nd cross check FPC
00001938  00000000 00000000                       1208+          DS    FD                     gap
00001940  00000000                                1209+          DS    F                   debug area
00001944  00000000                                1210+          DS    F
                                                  1211+*
00001948                                          1212+X8        DS    0F
00001948  B29D 852C                     0000072C  1213+          LFPC  FPCMOFF             initialize FPC
0000194C  E320 500C 0014                000018B4  1214+          LGF   R2,V2_8             get v2
00001952  E767 2000 0C36                00000000  1215+          VLM   V22,V23,0(R2)
00001958  E666 7002 0E75                          1216+          VCRNF V22,V22,V23,0,2         test instruction (dest is source)
0000195E  B29C 5040                     000018E8  1217+          STFPC FPC_R_8             save FPC
00001962  E760 5028 080E                000018D0  1218+          VST   V22,V1O8            save instruction result
00001968  07FB                                    1219+          BR    R11                 return
0000196C                                          1220+RE8       DS    0F                  expected 16 byte result
0000196C                                          1221+          DROP  R5
0000196C  00000000 00000000                       1222           DC    XL16'0000000000000000 8000000000000000'
00001974  80000000 00000000
0000197C  00800000 00000000                       1223           DC    XL16'0080000000000000 0000000000000000'
00001984  00000000 00000000
0000198C  80800000 00000000                       1224           DC    XL16'8080000000000000 0000000000000000'
00001994  00000000 00000000
                                                  1225
                                                  1226 * +2^(-149), -2^(-149) - subnormal
                                                  1227           VRR_C VCRNF,0,2,10,00,S                skip xc: underflow
000019A0                                          1228+          DS    0FD
000019A0                      000019A0            1229+          USING *,R5                base for test data and test routine
000019A0  00001A40                                1230+T9        DC    A(X9)               address of test routine
000019A4  0009                                    1231+          DC    H'9'                test number
000019A6  00                                      1232+          DC    X'00'
000019A7  E2                                      1233+          DC    CL1'S'              Y = skip cross check
000019A8  00                                      1234+          DC    HL1'0'              m4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000019A9  02                                       1235+         DC    HL1'2'            m5
000019AA  10                                       1236+FLG9     DC    X'10'             expected FPC flags
000019AB  00                                       1237+VXC9     DC    X'00'             expected VXC
000019AC  00001A74                                 1238+V2_9     DC    A(RE9+16)         address of v2: 16-byte packed decimal
000019B0  E5C3D9D5 C6404040                        1239+         DC    CL8'VCRNF'        instruction name
000019B8  00000010                                 1240+         DC    A(16)             result length
000019BC  00001A64                                 1241+         DC    A(RE9)            address of expected resul
000019C0  00000000 00000000                        1242+         DS    FD                gap
000019C8  00000000 00000000                        1243+V1O9     DS    XL16              V1 output
000019D0  00000000 00000000
000019D8  00000000 00000000                        1244+         DS    FD                   gap
000019E0  00000000                                 1245+FPC_R_9  DS    F                 FPC after instruction
000019E8  00000000 00000000                        1246+         DS    FD                   gap
000019F0  40404040 40404040                        1247+         DC    CL8' '            was cross check skipped?
000019F8  00000000 00000000                        1248+         DS    FD                   gap
00001A00  00000000 00000000                        1249+XCO9     DS    XL16              Cross check Output
00001A08  00000000 00000000
00001A10  00000000 00000000                        1250+         DS    XL16
00001A18  00000000 00000000
00001A20  00000000 00000000                        1251+         DS    FD                   gap
00001A28  00000000                                 1252+FPC_XC_9 DS    F                1st cross check FPC
00001A2C  00000000                                 1253+         DS    F                2nd cross check FPC
00001A30  00000000 00000000                        1254+         DS    FD                   gap
00001A38  00000000                                 1255+         DS    F                 debug area
00001A3C  00000000                                 1256+         DS    F
                                                   1257+*
00001A40                                           1258+X9       DS    0F
00001A40  B29D 852C                      0000072C  1259+         LFPC  FPCMOFF           initialize FPC
00001A44  E320 500C 0014                 000019AC  1260+         LGF   R2,V2_9           get v2
00001A4A  E767 2000 0C36                 00000000  1261+         VLM   V22,V23,0(R2)
00001A50  E666 7002 0E75                           1262+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001A56  B29C 5040                      000019E0  1263+         STFPC FPC_R_9           save FPC
00001A5A  E760 5028 080E                 000019C8  1264+         VST   V22,V1O9          save instruction result
00001A60  07FB                                     1265+         BR    R11               return
00001A64                                           1266+RE9      DS    0F                expected 16 byte result
00001A64                                           1267+         DROP  R5
00001A64  00000000 00000000                        1268          DC    XL16'0000000000000000 8000000000000000'
00001A6C  80000000 00000000
00001A74  00000001 00000000                        1269          DC    XL16'0000000100000000 0000000000000000'
00001A7C  00000000 00000000
00001A84  80800001 00000000                        1270          DC    XL16'8080000100000000 0000000000000000'
00001A8C  00000000 00000000
                                                   1271
                                                   1272 * +2^(128) * (1 - 2^(-24)) , - +2^(128) * (1 - 2^(-24)) : 3402823ï€¥10+32
                                                   1273          VRR_C VCRNF,0,2,20,00,S                skip xc: overflow
00001A98                                           1274+         DS    0FD
00001A98                       00001A98            1275+         USING *,R5              base for test data and test routine
00001A98  00001B38                                 1276+T10      DC    A(X10)            address of test routine
00001A9C  000A                                     1277+         DC    H'10'             test number
00001A9E  00                                       1278+         DC    X'00'
00001A9F  E2                                       1279+         DC    CL1'S'            Y = skip cross check
00001AA0  00                                       1280+         DC    HL1'0'            m4
00001AA1  02                                       1281+         DC    HL1'2'            m5
00001AA2  20                                       1282+FLG10    DC    X'20'             expected FPC flags
00001AA3  00                                       1283+VXC10    DC    X'00'             expected VXC
00001AA4  00001B6C                                 1284+V2_10    DC    A(RE10+16)        address of v2: 16-byte packed decimal
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00001AA8  E5C3D9D5 C6404040                     1285+         DC    CL8'VCRNF'         instruction name
00001AB0  00000010                              1286+         DC    A(16)              result length
00001AB4  00001B5C                              1287+         DC    A(RE10)            address of expected resul
00001AB8  00000000 00000000                     1288+         DS    FD                 gap
00001AC0  00000000 00000000                     1289+V1O10    DS    XL16               V1 output
00001AC8  00000000 00000000
00001AD0  00000000 00000000                     1290+         DS    FD                    gap
00001AD8  00000000                              1291+FPC_R_10 DS    F                  FPC after instruction
00001AE0  00000000 00000000                     1292+         DS    FD                    gap
00001AE8  40404040 40404040                     1293+         DC    CL8' '             was cross check skipped?
00001AF0  00000000 00000000                     1294+         DS    FD                    gap
00001AF8  00000000 00000000                     1295+XCO10    DS    XL16               Cross check Output
00001B00  00000000 00000000
00001B08  00000000 00000000                     1296+         DS    XL16
00001B10  00000000 00000000
00001B18  00000000 00000000                     1297+         DS    FD                    gap
00001B20  00000000                              1298+FPC_XC_10 DS   F                 1st cross check FPC
00001B24  00000000                              1299+         DS    F                 2nd cross check FPC
00001B28  00000000 00000000                     1300+         DS    FD                    gap
00001B30  00000000                              1301+         DS    F                  debug area
00001B34  00000000                              1302+         DS    F
                                                1303+*
00001B38                                        1304+X10      DS    0F
00001B38  B29D 852C                   0000072C  1305+         LFPC  FPCMOFF            initialize FPC
00001B3C  E320 500C 0014              00001AA4  1306+         LGF   R2,V2_10           get v2
00001B42  E767 2000 0C36              00000000  1307+         VLM   V22,V23,0(R2)
00001B48  E666 7002 0E75                        1308+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00001B4E  B29C 5040                   00001AD8  1309+         STFPC FPC_R_10           save FPC
00001B52  E760 5028 080E              00001AC0  1310+         VST   V22,V1O10          save instruction result
00001B58  07FB                                  1311+         BR    R11                return
00001B5C                                        1312+RE10     DS    0F                 expected 16 byte result
00001B5C                                        1313+         DROP  R5
00001B5C  7FFF0000 00000000                     1314          DC    XL16'7FFF000000000000 FFFF000000000000'
00001B64  FFFF0000 00000000
00001B6C  7F7FFFFF 00000000                     1315          DC    XL16'7F7FFFFF00000000 0000000000000000'
00001B74  00000000 00000000
00001B7C  FF7FFFFF 00000000                     1316          DC    XL16'FF7FFFFF00000000 0000000000000000'
00001B84  00000000 00000000
                                                1317
                                                1318 * NAN, -NAN
                                                1319          VRR_C VCRNF,0,2,80,00,S                skip xc: invalid on XC
00001B90                                        1320+         DS    0FD
00001B90                    00001B90            1321+         USING *,R5               base for test data and test routine
00001B90  00001C30                              1322+T11      DC    A(X11)             address of test routine
00001B94  000B                                  1323+         DC    H'11'              test number
00001B96  00                                    1324+         DC    X'00'
00001B97  E2                                    1325+         DC    CL1'S'             Y = skip cross check
00001B98  00                                    1326+         DC    HL1'0'             m4
00001B99  02                                    1327+         DC    HL1'2'             m5
00001B9A  80                                    1328+FLG11    DC    X'80'              expected FPC flags
00001B9B  00                                    1329+VXC11    DC    X'00'              expected VXC
00001B9C  00001C64                              1330+V2_11    DC    A(RE11+16)         address of v2: 16-byte packed decimal
00001BA0  E5C3D9D5 C6404040                     1331+         DC    CL8'VCRNF'         instruction name
00001BA8  00000010                              1332+         DC    A(16)              result length
00001BAC  00001C54                              1333+         DC    A(RE11)            address of expected resul
00001BB0  00000000 00000000                     1334+         DS    FD                 gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001BB8  00000000 00000000                        1335+V1O11    DS    XL16              V1 output
00001BC0  00000000 00000000
00001BC8  00000000 00000000                        1336+         DS    FD                   gap
00001BD0  00000000                                 1337+FPC_R_11 DS    F                 FPC after instruction
00001BD8  00000000 00000000                        1338+         DS    FD                   gap
00001BE0  40404040 40404040                        1339+         DC    CL8' '            was cross check skipped?
00001BE8  00000000 00000000                        1340+         DS    FD                   gap
00001BF0  00000000 00000000                        1341+XCO11    DS    XL16              Cross check Output
00001BF8  00000000 00000000
00001C00  00000000 00000000                        1342+         DS    XL16
00001C08  00000000 00000000
00001C10  00000000 00000000                        1343+         DS    FD                   gap
00001C18  00000000                                 1344+FPC_XC_11 DS   F                1st cross check FPC
00001C1C  00000000                                 1345+         DS    F                2nd cross check FPC
00001C20  00000000 00000000                        1346+         DS    FD                   gap
00001C28  00000000                                 1347+         DS    F                 debug area
00001C2C  00000000                                 1348+         DS    F
                                                   1349+*
00001C30                                           1350+X11      DS    0F
00001C30  B29D 852C                      0000072C  1351+         LFPC  FPCMOFF           initialize FPC
00001C34  E320 500C 0014                 00001B9C  1352+         LGF   R2,V2_11          get v2
00001C3A  E767 2000 0C36                 00000000  1353+         VLM   V22,V23,0(R2)
00001C40  E666 7002 0E75                           1354+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001C46  B29C 5040                      00001BD0  1355+         STFPC FPC_R_11          save FPC
00001C4A  E760 5028 080E                 00001BB8  1356+         VST   V22,V1O11         save instruction result
00001C50  07FB                                     1357+         BR    R11               return
00001C54                                           1358+RE11     DS    0F                expected 16 byte result
00001C54                                           1359+         DROP  R5
00001C54  7FFF0000 00000000                        1360          DC    XL16'7FFF000000000000 FFFF000000000000'
00001C5C  FFFF0000 00000000
00001C64  7FC00000 00000000                        1361          DC    XL16'7FC0000000000000 0000000000000000'
00001C6C  00000000 00000000
00001C74  FFC00000 00000000                        1362          DC    XL16'FFC0000000000000 0000000000000000'
00001C7C  00000000 00000000
                                                   1363
                                                   1364 * bad m4 - inexact
                                                   1365          VRR_C VCRNF,2,2,08,00,S                skip xc: inexact
00001C88                                           1366+         DS    0FD
00001C88                       00001C88            1367+         USING *,R5              base for test data and test routine
00001C88  00001D28                                 1368+T12      DC    A(X12)            address of test routine
00001C8C  000C                                     1369+         DC    H'12'             test number
00001C8E  00                                       1370+         DC    X'00'
00001C8F  E2                                       1371+         DC    CL1'S'            Y = skip cross check
00001C90  02                                       1372+         DC    HL1'2'            m4
00001C91  02                                       1373+         DC    HL1'2'            m5
00001C92  08                                       1374+FLG12    DC    X'08'             expected FPC flags
00001C93  00                                       1375+VXC12    DC    X'00'             expected VXC
00001C94  00001D5C                                 1376+V2_12    DC    A(RE12+16)        address of v2: 16-byte packed decimal
00001C98  E5C3D9D5 C6404040                        1377+         DC    CL8'VCRNF'        instruction name
00001CA0  00000010                                 1378+         DC    A(16)             result length
00001CA4  00001D4C                                 1379+         DC    A(RE12)           address of expected resul
00001CA8  00000000 00000000                        1380+         DS    FD                gap
00001CB0  00000000 00000000                        1381+V1O12    DS    XL16              V1 output
00001CB8  00000000 00000000
00001CC0  00000000 00000000                        1382+         DS    FD                   gap
00001CC8  00000000                                 1383+FPC_R_12 DS    F                 FPC after instruction
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001CD0  00000000 00000000                      1384+          DS    FD                     gap
00001CD8  40404040 40404040                      1385+          DC    CL8' '              was cross check skipped?
00001CE0  00000000 00000000                      1386+          DS    FD                     gap
00001CE8  00000000 00000000                      1387+XC012     DS    XL16                Cross check Output
00001CF0  00000000 00000000
00001CF8  00000000 00000000                      1388+          DS    XL16
00001D00  00000000 00000000
00001D08  00000000 00000000                      1389+          DS    FD                     gap
00001D10  00000000                               1390+FPC_XC_12 DS    F                  1st cross check FPC
00001D14  00000000                               1391+          DS    F                  2nd cross check FPC
00001D18  00000000 00000000                      1392+          DS    FD                     gap
00001D20  00000000                               1393+          DS    F                   debug area
00001D24  00000000                               1394+          DS    F
                                                 1395+*
00001D28                                         1396+X12       DS    0F
00001D28  B29D 852C                    0000072C  1397+          LFPC  FPCMOFF             initialize FPC
00001D2C  E320 500C 0014               00001C94  1398+          LGF   R2,V2_12            get v2
00001D32  E767 2000 0C36               00000000  1399+          VLM   V22,V23,0(R2)
00001D38  E666 7002 2E75                         1400+          VCRNF V22,V22,V23,2,2         test instruction (dest is source)
00001D3E  B29C 5040                    00001CC8  1401+          STFPC FPC_R_12            save FPC
00001D42  E760 5028 080E               00001CB0  1402+          VST   V22,V1O12           save instruction result
00001D48  07FB                                   1403+          BR    R11                 return
00001D4C                                         1404+RE12      DS    0F                  expected 16 byte result
00001D4C                                         1405+          DROP  R5
00001D4C  00000000 00000000                      1406           DC    XL16'0000000000000000 0000000000000000'
00001D54  00000000 00000000
00001D5C  7FC00000 00000000                      1407           DC    XL16'7FC0000000000000 0000000000000000'
00001D64  00000000 00000000
00001D6C  FFC00000 00000000                      1408           DC    XL16'FFC0000000000000 0000000000000000'
00001D74  00000000 00000000
                                                 1409
                                                 1410 * bad m5 - inexact
                                                 1411           VRR_C VCRNF,0,3,08,00,S                skip xc: inexact
00001D80                                         1412+          DS    0FD
00001D80                     00001D80            1413+          USING *,R5                base for test data and test routine
00001D80  00001E20                               1414+T13       DC    A(X13)              address of test routine
00001D84  000D                                   1415+          DC    H'13'               test number
00001D86  00                                     1416+          DC    X'00'
00001D87  E2                                     1417+          DC    CL1'S'              Y = skip cross check
00001D88  00                                     1418+          DC    HL1'0'              m4
00001D89  03                                     1419+          DC    HL1'3'              m5
00001D8A  08                                     1420+FLG13     DC    X'08'               expected FPC flags
00001D8B  00                                     1421+VXC13     DC    X'00'               expected VXC
00001D8C  00001E54                               1422+V2_13     DC    A(RE13+16)          address of v2: 16-byte packed decimal
00001D90  E5C3D9D5 C6404040                      1423+          DC    CL8'VCRNF'          instruction name
00001D98  00000010                               1424+          DC    A(16)               result length
00001D9C  00001E44                               1425+          DC    A(RE13)             address of expected resul
00001DA0  00000000 00000000                      1426+          DS    FD                  gap
00001DA8  00000000 00000000                      1427+V1O13     DS    XL16                V1 output
00001DB0  00000000 00000000
00001DB8  00000000 00000000                      1428+          DS    FD                     gap
00001DC0  00000000                               1429+FPC_R_13  DS    F                   FPC after instruction
00001DC8  00000000 00000000                      1430+          DS    FD                     gap
00001DD0  40404040 40404040                      1431+          DC    CL8' '              was cross check skipped?
00001DD8  00000000 00000000                      1432+          DS    FD                     gap
00001DE0  00000000 00000000                      1433+XC013     DS    XL16                Cross check Output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001DE8  00000000 00000000
00001DF0  00000000 00000000                       1434+          DS    XL16
00001DF8  00000000 00000000
00001E00  00000000 00000000                       1435+          DS    FD                     gap
00001E08  00000000                                1436+FPC_XC_13 DS    F                  1st cross check FPC
00001E0C  00000000                                1437+          DS    F                  2nd cross check FPC
00001E10  00000000 00000000                       1438+          DS    FD                     gap
00001E18  00000000                                1439+          DS    F                   debug area
00001E1C  00000000                                1440+          DS    F
                                                  1441+*
00001E20                                          1442+X13       DS    0F
00001E20  B29D 852C                     0000072C  1443+          LFPC  FPCMOFF             initialize FPC
00001E24  E320 500C 0014                00001D8C  1444+          LGF   R2,V2_13            get v2
00001E2A  E767 2000 0C36                00000000  1445+          VLM   V22,V23,0(R2)
00001E30  E666 7003 0E75                          1446+          VCRNF V22,V22,V23,0,3       test instruction (dest is source)
00001E36  B29C 5040                     00001DC0  1447+          STFPC FPC_R_13            save FPC
00001E3A  E760 5028 080E                00001DA8  1448+          VST   V22,V1O13           save instruction result
00001E40  07FB                                    1449+          BR    R11                 return
00001E44                                          1450+RE13      DS    0F                  expected 16 byte result
00001E44                                          1451+          DROP  R5
00001E44  00000000 00000000                       1452           DC    XL16'0000000000000000 0000000000000000'
00001E4C  00000000 00000000
00001E54  7FC00000 00000000                       1453           DC    XL16'7FC0000000000000 0000000000000000'
00001E5C  00000000 00000000
00001E64  FFC00000 00000000                       1454           DC    XL16'FFC0000000000000 0000000000000000'
00001E6C  00000000 00000000
                                                  1455
                                                  1456 * Additional tests
                                                  1457 *-------------------------------------------------------------------
                                                  1458 * +99.1, -99.1
                                                  1459           VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00001E78                                          1460+          DS    0FD
00001E78                       00001E78           1461+          USING *,R5              base for test data and test routine
00001E78  00001F18                                1462+T14       DC    A(X14)              address of test routine
00001E7C  000E                                    1463+          DC    H'14'               test number
00001E7E  00                                      1464+          DC    X'00'
00001E7F  E2                                      1465+          DC    CL1'S'              Y = skip cross check
00001E80  00                                      1466+          DC    HL1'0'              m4
00001E81  02                                      1467+          DC    HL1'2'              m5
00001E82  00                                      1468+FLG14     DC    X'00'               expected FPC flags
00001E83  00                                      1469+VXC14     DC    X'00'               expected VXC
00001E84  00001F4C                                1470+V2_14     DC    A(RE14+16)          address of v2: 16-byte packed decimal
00001E88  E5C3D9D5 C6404040                       1471+          DC    CL8'VCRNF'          instruction name
00001E90  00000010                                1472+          DC    A(16)               result length
00001E94  00001F3C                                1473+          DC    A(RE14)             address of expected resul
00001E98  00000000 00000000                       1474+          DS    FD                  gap
00001EA0  00000000 00000000                       1475+V1O14     DS    XL16                V1 output
00001EA8  00000000 00000000
00001EB0  00000000 00000000                       1476+          DS    FD                     gap
00001EB8  00000000                                1477+FPC_R_14  DS    F                   FPC after instruction
00001EC0  00000000 00000000                       1478+          DS    FD                     gap
00001EC8  40404040 40404040                       1479+          DC    CL8' '              was cross check skipped?
00001ED0  00000000 00000000                       1480+          DS    FD                     gap
00001ED8  00000000 00000000                       1481+XCO14     DS    XL16                Cross check Output
00001EE0  00000000 00000000
00001EE8  00000000 00000000                       1482+          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001EF0  00000000 00000000
00001EF8  00000000 00000000                        1483+          DS    FD                      gap
00001F00  00000000                                 1484+FPC_XC_14 DS    F                   1st cross check FPC
00001F04  00000000                                 1485+          DS    F                   2nd cross check FPC
00001F08  00000000 00000000                        1486+          DS    FD                      gap
00001F10  00000000                                 1487+          DS    F                    debug area
00001F14  00000000                                 1488+          DS    F
                                                   1489+*
00001F18                                           1490+X14       DS    0F
00001F18  B29D 852C                      0000072C  1491+          LFPC  FPCMOFF              initialize FPC
00001F1C  E320 500C 0014                 00001E84  1492+          LGF   R2,V2_14             get v2
00001F22  E767 2000 0C36                 00000000  1493+          VLM   V22,V23,0(R2)
00001F28  E666 7002 0E75                           1494+          VCRNF V22,V22,V23,0,2          test instruction (dest is source)
00001F2E  B29C 5040                      00001EB8  1495+          STFPC FPC_R_14             save FPC
00001F32  E760 5028 080E                 00001EA0  1496+          VST   V22,V1O14            save instruction result
00001F38  07FB                                     1497+          BR    R11                  return
00001F3C                                           1498+RE14      DS    0F                   expected 16 byte result
00001F3C                                           1499+          DROP  R5
00001F3C  4B190000 00000000                        1500           DC    XL16'4B19000000000000 CB19000000000000'
00001F44  CB190000 00000000
00001F4C  42C63333 00000000                        1501           DC    XL16'42C6333300000000 0000000000000000'
00001F54  00000000 00000000
00001F5C  C2C63333 00000000                        1502           DC    XL16'C2C6333300000000 0000000000000000'
00001F64  00000000 00000000
                                                   1503
                                                   1504 * +100.9, -100.9
                                                   1505           VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00001F70                                           1506+          DS    0FD
00001F70                       00001F70            1507+          USING *,R5                 base for test data and test routine
00001F70  00002010                                 1508+T15       DC    A(X15)               address of test routine
00001F74  000F                                     1509+          DC    H'15'                test number
00001F76  00                                       1510+          DC    X'00'
00001F77  E2                                       1511+          DC    CL1'S'               Y = skip cross check
00001F78  00                                       1512+          DC    HL1'0'               m4
00001F79  02                                       1513+          DC    HL1'2'               m5
00001F7A  00                                       1514+FLG15     DC    X'00'                expected FPC flags
00001F7B  00                                       1515+VXC15     DC    X'00'                expected VXC
00001F7C  00002044                                 1516+V2_15     DC    A(RE15+16)           address of v2: 16-byte packed decimal
00001F80  E5C3D9D5 C6404040                        1517+          DC    CL8'VCRNF'           instruction name
00001F88  00000010                                 1518+          DC    A(16)                result length
00001F8C  00002034                                 1519+          DC    A(RE15)              address of expected resul
00001F90  00000000 00000000                        1520+          DS    FD                   gap
00001F98  00000000 00000000                        1521+V1O15     DS    XL16                 V1 output
00001FA0  00000000 00000000
00001FA8  00000000 00000000                        1522+          DS    FD                      gap
00001FB0  00000000                                 1523+FPC_R_15  DS    F                    FPC after instruction
00001FB8  00000000 00000000                        1524+          DS    FD                      gap
00001FC0  40404040 40404040                        1525+          DC    CL8' '               was cross check skipped?
00001FC8  00000000 00000000                        1526+          DS    FD                      gap
00001FD0  00000000 00000000                        1527+XCO15     DS    XL16                 Cross check Output
00001FD8  00000000 00000000
00001FE0  00000000 00000000                        1528+          DS    XL16
00001FE8  00000000 00000000
00001FF0  00000000 00000000                        1529+          DS    FD                      gap
00001FF8  00000000                                 1530+FPC_XC_15 DS    F                   1st cross check FPC
00001FFC  00000000                                 1531+          DS    F                   2nd cross check FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002000  00000000 00000000                       1532+         DS    FD                    gap
00002008  00000000                                1533+         DS    F                  debug area
0000200C  00000000                                1534+         DS    F
                                                  1535+*
00002010                                          1536+X15      DS    0F
00002010  B29D 852C                     0000072C  1537+         LFPC  FPCMOFF            initialize FPC
00002014  E320 500C 0014                00001F7C  1538+         LGF   R2,V2_15           get v2
0000201A  E767 2000 0C36                00000000  1539+         VLM   V22,V23,0(R2)
00002020  E666 7002 0E75                          1540+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00002026  B29C 5040                     00001FB0  1541+         STFPC FPC_R_15           save FPC
0000202A  E760 5028 080E                00001F98  1542+         VST   V22,V1O15          save instruction result
00002030  07FB                                    1543+         BR    R11                return
00002034                                          1544+RE15     DS    0F                 expected 16 byte result
00002034                                          1545+         DROP  R5
00002034  4B270000 00000000                       1546          DC    XL16'4B27000000000000 CB27000000000000'
0000203C  CB270000 00000000
00002044  42C9CCCD 00000000                       1547          DC    XL16'42C9CCCD00000000 0000000000000000'
0000204C  00000000 00000000
00002054  C2C9CCCD 00000000                       1548          DC    XL16'C2C9CCCD00000000 0000000000000000'
0000205C  00000000 00000000
                                                  1549
                                                  1550 * +527.7895, -527.7895
                                                  1551          VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00002068                                          1552+         DS    0FD
00002068                      00002068            1553+         USING *,R5               base for test data and test routine
00002068  00002108                                1554+T16      DC    A(X16)             address of test routine
0000206C  0010                                    1555+         DC    H'16'              test number
0000206E  00                                      1556+         DC    X'00'
0000206F  E2                                      1557+         DC    CL1'S'             Y = skip cross check
00002070  00                                      1558+         DC    HL1'0'             m4
00002071  02                                      1559+         DC    HL1'2'             m5
00002072  00                                      1560+FLG16    DC    X'00'              expected FPC flags
00002073  00                                      1561+VXC16    DC    X'00'              expected VXC
00002074  0000213C                                1562+V2_16    DC    A(RE16+16)         address of v2: 16-byte packed decimal
00002078  E5C3D9D5 C6404040                       1563+         DC    CL8'VCRNF'         instruction name
00002080  00000010                                1564+         DC    A(16)              result length
00002084  0000212C                                1565+         DC    A(RE16)            address of expected resul
00002088  00000000 00000000                       1566+         DS    FD                 gap
00002090  00000000 00000000                       1567+V1O16    DS    XL16               V1 output
00002098  00000000 00000000
000020A0  00000000 00000000                       1568+         DS    FD                    gap
000020A8  00000000                                1569+FPC_R_16 DS    F                  FPC after instruction
000020B0  00000000 00000000                       1570+         DS    FD                    gap
000020B8  40404040 40404040                       1571+         DC    CL8' '             was cross check skipped?
000020C0  00000000 00000000                       1572+         DS    FD                    gap
000020C8  00000000 00000000                       1573+XCO16    DS    XL16               Cross check Output
000020D0  00000000 00000000
000020D8  00000000 00000000                       1574+         DS    XL16
000020E0  00000000 00000000
000020E8  00000000 00000000                       1575+         DS    FD                    gap
000020F0  00000000                                1576+FPC_XC_16 DS   F                 1st cross check FPC
000020F4  00000000                                1577+         DS    F                 2nd cross check FPC
000020F8  00000000 00000000                       1578+         DS    FD                    gap
00002100  00000000                                1579+         DS    F                  debug area
00002104  00000000                                1580+         DS    F
                                                  1581+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002108                                         1582+X16      DS    0F
00002108  B29D 852C                     0000072C 1583+         LFPC  FPCMOFF            initialize FPC
0000210C  E320 500C 0014                00002074 1584+         LGF   R2,V2_16           get v2
00002112  E767 2000 0C36                00000000 1585+         VLM   V22,V23,0(R2)
00002118  E666 7002 0E75                         1586+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
0000211E  B29C 5040                     000020A8 1587+         STFPC FPC_R_16           save FPC
00002122  E760 5028 080E                00002090 1588+         VST   V22,V1O16          save instruction result
00002128  07FB                                   1589+         BR    R11                return
0000212C                                         1590+RE16     DS    0F                 expected 16 byte result
0000212C                                         1591+         DROP  R5
0000212C  50100000 00000000                      1592          DC    XL16'5010000000000000 D010000000000000'
00002134  D0100000 00000000
0000213C  4403F287 00000000                      1593          DC    XL16'4403F28700000000 0000000000000000'
00002144  00000000 00000000
0000214C  C403F287 00000000                      1594          DC    XL16'C403F28700000000 0000000000000000'
00002154  00000000 00000000
                                                 1595
                                                 1596 * +1055.5790, -1055.5790
                                                 1597          VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00002160                                         1598+         DS    0FD
00002160                      00002160           1599+         USING *,R5               base for test data and test routine
00002160  00002200                               1600+T17      DC    A(X17)             address of test routine
00002164  0011                                   1601+         DC    H'17'              test number
00002166  00                                     1602+         DC    X'00'
00002167  E2                                     1603+         DC    CL1'S'             Y = skip cross check
00002168  00                                     1604+         DC    HL1'0'             m4
00002169  02                                     1605+         DC    HL1'2'             m5
0000216A  00                                     1606+FLG17    DC    X'00'              expected FPC flags
0000216B  00                                     1607+VXC17    DC    X'00'              expected VXC
0000216C  00002234                               1608+V2_17    DC    A(RE17+16)         address of v2: 16-byte packed decimal
00002170  E5C3D9D5 C6404040                      1609+         DC    CL8'VCRNF'         instruction name
00002178  00000010                               1610+         DC    A(16)              result length
0000217C  00002224                               1611+         DC    A(RE17)            address of expected resul
00002180  00000000 00000000                      1612+         DS    FD                 gap
00002188  00000000 00000000                      1613+V1O17    DS    XL16               V1 output
00002190  00000000 00000000
00002198  00000000 00000000                      1614+         DS    FD                    gap
000021A0  00000000                               1615+FPC_R_17 DS    F                  FPC after instruction
000021A8  00000000 00000000                      1616+         DS    FD                    gap
000021B0  40404040 40404040                      1617+         DC    CL8' '             was cross check skipped?
000021B8  00000000 00000000                      1618+         DS    FD                    gap
000021C0  00000000 00000000                      1619+XCO17    DS    XL16               Cross check Output
000021C8  00000000 00000000
000021D0  00000000 00000000                      1620+         DS    XL16
000021D8  00000000 00000000
000021E0  00000000 00000000                      1621+         DS    FD                    gap
000021E8  00000000                               1622+FPC_XC_17 DS   F                 1st cross check FPC
000021EC  00000000                               1623+         DS    F                 2nd cross check FPC
000021F0  00000000 00000000                      1624+         DS    FD                    gap
000021F8  00000000                               1625+         DS    F                  debug area
000021FC  00000000                               1626+         DS    F
                                                 1627+*
00002200                                         1628+X17      DS    0F
00002200  B29D 852C                     0000072C 1629+         LFPC  FPCMOFF            initialize FPC
00002204  E320 500C 0014                0000216C 1630+         LGF   R2,V2_17           get v2
0000220A  E767 2000 0C36                00000000 1631+         VLM   V22,V23,0(R2)
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00002210  E666 7002 0E75                          1632+          VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002216  B29C 5040                     000021A0  1633+          STFPC FPC_R_17          save FPC
0000221A  E760 5028 080E                00002188  1634+          VST   V22,V1O17         save instruction result
00002220  07FB                                    1635+          BR    R11               return
00002224                                          1636+RE17      DS    0F                expected 16 byte result
00002224                                          1637+          DROP  R5
00002224  52100000 00000000                       1638           DC    XL16'5210000000000000 D210000000000000'
0000222C  D2100000 00000000
00002234  4483F287 00000000                       1639           DC    XL16'4483F28700000000 0000000000000000'
0000223C  00000000 00000000
00002244  C483F287 00000000                       1640           DC    XL16'C483F28700000000 0000000000000000'
0000224C  00000000 00000000
                                                  1641
                                                  1642 * +Zero, -Zero
                                                  1643           VRR_C VCRNF,0,2,00,00,N
00002258                                          1644+          DS    0FD
00002258                      00002258            1645+          USING *,R5              base for test data and test routine
00002258  000022F8                                1646+T18       DC    A(X18)            address of test routine
0000225C  0012                                    1647+          DC    H'18'             test number
0000225E  00                                      1648+          DC    X'00'
0000225F  D5                                      1649+          DC    CL1'N'            Y = skip cross check
00002260  00                                      1650+          DC    HL1'0'            m4
00002261  02                                      1651+          DC    HL1'2'            m5
00002262  00                                      1652+FLG18     DC    X'00'             expected FPC flags
00002263  00                                      1653+VXC18     DC    X'00'             expected VXC
00002264  0000232C                                1654+V2_18     DC    A(RE18+16)        address of v2: 16-byte packed decimal
00002268  E5C3D9D5 C6404040                       1655+          DC    CL8'VCRNF'        instruction name
00002270  00000010                                1656+          DC    A(16)             result length
00002274  0000231C                                1657+          DC    A(RE18)           address of expected resul
00002278  00000000 00000000                       1658+          DS    FD                gap
00002280  00000000 00000000                       1659+V1O18     DS    XL16              V1 output
00002288  00000000 00000000
00002290  00000000 00000000                       1660+          DS    FD                   gap
00002298  00000000                                1661+FPC_R_18  DS    F                 FPC after instruction
000022A0  00000000 00000000                       1662+          DS    FD                   gap
000022A8  40404040 40404040                       1663+          DC    CL8' '            was cross check skipped?
000022B0  00000000 00000000                       1664+          DS    FD                   gap
000022B8  00000000 00000000                       1665+XCO18     DS    XL16              Cross check Output
000022C0  00000000 00000000
000022C8  00000000 00000000                       1666+          DS    XL16
000022D0  00000000 00000000
000022D8  00000000 00000000                       1667+          DS    FD                   gap
000022E0  00000000                                1668+FPC_XC_18 DS    F                1st cross check FPC
000022E4  00000000                                1669+          DS    F                2nd cross check FPC
000022E8  00000000 00000000                       1670+          DS    FD                   gap
000022F0  00000000                                1671+          DS    F                 debug area
000022F4  00000000                                1672+          DS    F
                                                  1673+*
000022F8                                          1674+X18       DS    0F
000022F8  B29D 852C                     0000072C  1675+          LFPC  FPCMOFF           initialize FPC
000022FC  E320 500C 0014                00002264  1676+          LGF   R2,V2_18          get v2
00002302  E767 2000 0C36                00000000  1677+          VLM   V22,V23,0(R2)
00002308  E666 7002 0E75                          1678+          VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000230E  B29C 5040                     00002298  1679+          STFPC FPC_R_18          save FPC
00002312  E760 5028 080E                00002280  1680+          VST   V22,V1O18         save instruction result
00002318  07FB                                    1681+          BR    R11               return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000231C                                           1682+RE18     DS    0F                  expected 16 byte result
0000231C                                           1683+         DROP  R5
0000231C  00000000 00000000                        1684          DC    XL16'0000000000000000 8000000000000000'
00002324  80000000 00000000
0000232C  00000000 00000000                        1685          DC    XL16'0000000000000000 0000000000000000'
00002334  00000000 00000000
0000233C  80000000 00000000                        1686          DC    XL16'8000000000000000 0000000000000000'
00002344  00000000 00000000
                                                   1687
                                                   1688 * +Normal, -Normal
                                                   1689          VRR_C VCRNF,0,2,00,00,N                skip xc: lost digits
00002350                                           1690+         DS    0FD
00002350                     00002350              1691+         USING *,R5                base for test data and test routine
00002350  000023F0                                 1692+T19      DC    A(X19)              address of test routine
00002354  0013                                     1693+         DC    H'19'               test number
00002356  00                                       1694+         DC    X'00'
00002357  D5                                       1695+         DC    CL1'N'              Y = skip cross check
00002358  00                                       1696+         DC    HL1'0'              m4
00002359  02                                       1697+         DC    HL1'2'              m5
0000235A  00                                       1698+FLG19    DC    X'00'               expected FPC flags
0000235B  00                                       1699+VXC19    DC    X'00'               expected VXC
0000235C  00002424                                 1700+V2_19    DC    A(RE19+16)          address of v2: 16-byte packed decimal
00002360  E5C3D9D5 C6404040                        1701+         DC    CL8'VCRNF'          instruction name
00002368  00000010                                 1702+         DC    A(16)               result length
0000236C  00002414                                 1703+         DC    A(RE19)             address of expected resul
00002370  00000000 00000000                        1704+         DS    FD                  gap
00002378  00000000 00000000                        1705+V1O19    DS    XL16                V1 output
00002380  00000000 00000000
00002388  00000000 00000000                        1706+         DS    FD                     gap
00002390  00000000                                 1707+FPC_R_19 DS    F                   FPC after instruction
00002398  00000000 00000000                        1708+         DS    FD                     gap
000023A0  40404040 40404040                        1709+         DC    CL8' '              was cross check skipped?
000023A8  00000000 00000000                        1710+         DS    FD                     gap
000023B0  00000000 00000000                        1711+XCO19    DS    XL16                Cross check Output
000023B8  00000000 00000000
000023C0  00000000 00000000                        1712+         DS    XL16
000023C8  00000000 00000000
000023D0  00000000 00000000                        1713+         DS    FD                     gap
000023D8  00000000                                 1714+FPC_XC_19 DS   F                  1st cross check FPC
000023DC  00000000                                 1715+         DS    F                  2nd cross check FPC
000023E0  00000000 00000000                        1716+         DS    FD                     gap
000023E8  00000000                                 1717+         DS    F                   debug area
000023EC  00000000                                 1718+         DS    F
                                                   1719+*
000023F0                                           1720+X19      DS    0F
000023F0  B29D 852C                    0000072C    1721+         LFPC  FPCMOFF             initialize FPC
000023F4  E320 500C 0014               0000235C    1722+         LGF   R2,V2_19            get v2
000023FA  E767 2000 0C36               00000000    1723+         VLM   V22,V23,0(R2)
00002400  E666 7002 0E75                           1724+         VCRNF V22,V22,V23,0,2         test instruction (dest is source)
00002406  B29C 5040                    00002390    1725+         STFPC FPC_R_19            save FPC
0000240A  E760 5028 080E               00002378    1726+         VST   V22,V1O19           save instruction result
00002410  07FB                                     1727+         BR    R11                 return
00002414                                           1728+RE19     DS    0F                  expected 16 byte result
00002414                                           1729+         DROP  R5
00002414  40800000 00000000                        1730          DC    XL16'4080000000000000 C080000000000000'
0000241C  C0800000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002424  40200000 00000000                        1731          DC    XL16'4020000000000000 0000000000000000'
0000242C  00000000 00000000
00002434  C0200000 00000000                        1732          DC    XL16'C020000000000000 0000000000000000'
0000243C  00000000 00000000
                                                   1733
                                                   1734 * +Subnormal, -Subnormal
                                                   1735          VRR_C VCRNF,0,2,10,00,S                 skip xc: lost digits
00002448                                           1736+         DS    0FD
00002448                     00002448              1737+         USING *,R5              base for test data and test routine
00002448  000024E8                                 1738+T20      DC    A(X20)            address of test routine
0000244C  0014                                     1739+         DC    H'20'             test number
0000244E  00                                       1740+         DC    X'00'
0000244F  E2                                       1741+         DC    CL1'S'            Y = skip cross check
00002450  00                                       1742+         DC    HL1'0'            m4
00002451  02                                       1743+         DC    HL1'2'            m5
00002452  10                                       1744+FLG20    DC    X'10'             expected FPC flags
00002453  00                                       1745+VXC20    DC    X'00'             expected VXC
00002454  0000251C                                 1746+V2_20    DC    A(RE20+16)        address of v2: 16-byte packed decimal
00002458  E5C3D9D5 C6404040                        1747+         DC    CL8'VCRNF'        instruction name
00002460  00000010                                 1748+         DC    A(16)             result length
00002464  0000250C                                 1749+         DC    A(RE20)           address of expected resul
00002468  00000000 00000000                        1750+         DS    FD                gap
00002470  00000000 00000000                        1751+V1O20    DS    XL16              V1 output
00002478  00000000 00000000
00002480  00000000 00000000                        1752+         DS    FD                   gap
00002488  00000000                                 1753+FPC_R_20 DS    F                 FPC after instruction
00002490  00000000 00000000                        1754+         DS    FD                   gap
00002498  40404040 40404040                        1755+         DC    CL8' '            was cross check skipped?
000024A0  00000000 00000000                        1756+         DS    FD                   gap
000024A8  00000000 00000000                        1757+XCO20    DS    XL16              Cross check Output
000024B0  00000000 00000000
000024B8  00000000 00000000                        1758+         DS    XL16
000024C0  00000000 00000000
000024C8  00000000 00000000                        1759+         DS    FD                   gap
000024D0  00000000                                 1760+FPC_XC_20 DS   F                1st cross check FPC
000024D4  00000000                                 1761+         DS    F                2nd cross check FPC
000024D8  00000000 00000000                        1762+         DS    FD                   gap
000024E0  00000000                                 1763+         DS    F                 debug area
000024E4  00000000                                 1764+         DS    F
                                                   1765+*
000024E8                                           1766+X20      DS    0F
000024E8  B29D 852C                    0000072C    1767+         LFPC  FPCMOFF           initialize FPC
000024EC  E320 500C 0014               00002454    1768+         LGF   R2,V2_20          get v2
000024F2  E767 2000 0C36               00000000    1769+         VLM   V22,V23,0(R2)
000024F8  E666 7002 0E75                           1770+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000024FE  B29C 5040                    00002488    1771+         STFPC FPC_R_20          save FPC
00002502  E760 5028 080E               00002470    1772+         VST   V22,V1O20         save instruction result
00002508  07FB                                     1773+         BR    R11               return
0000250C                                           1774+RE20     DS    0F                expected 16 byte result
0000250C                                           1775+         DROP  R5
0000250C  00000000 00000000                        1776          DC    XL16'0000000000000000 8000000000000000'
00002514  80000000 00000000
0000251C  007FFFFF 00000000                        1777          DC    XL16'007FFFFF00000000 0000000000000000'
00002524  00000000 00000000
0000252C  807FFFFF 00000000                        1778          DC    XL16'807FFFFF00000000 0000000000000000'
00002534  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1779
                                                   1780 * +Infinity, -Infinity
                                                   1781          VRR_C VCRNF,0,2,20,00,S                  skip xc: lost digits
00002540                                           1782+         DS    0FD
00002540                        00002540           1783+         USING *,R5               base for test data and test routine
00002540  000025E0                                 1784+T21      DC    A(X21)             address of test routine
00002544  0015                                     1785+         DC    H'21'              test number
00002546  00                                       1786+         DC    X'00'
00002547  E2                                       1787+         DC    CL1'S'             Y = skip cross check
00002548  00                                       1788+         DC    HL1'0'             m4
00002549  02                                       1789+         DC    HL1'2'             m5
0000254A  20                                       1790+FLG21    DC    X'20'              expected FPC flags
0000254B  00                                       1791+VXC21    DC    X'00'              expected VXC
0000254C  00002614                                 1792+V2_21    DC    A(RE21+16)         address of v2: 16-byte packed decimal
00002550  E5C3D9D5 C6404040                        1793+         DC    CL8'VCRNF'         instruction name
00002558  00000010                                 1794+         DC    A(16)              result length
0000255C  00002604                                 1795+         DC    A(RE21)            address of expected resul
00002560  00000000 00000000                        1796+         DS    FD                 gap
00002568  00000000 00000000                        1797+V1O21    DS    XL16               V1 output
00002570  00000000 00000000
00002578  00000000 00000000                        1798+         DS    FD                    gap
00002580  00000000                                 1799+FPC_R_21 DS    F                  FPC after instruction
00002588  00000000 00000000                        1800+         DS    FD                    gap
00002590  40404040 40404040                        1801+         DC    CL8' '             was cross check skipped?
00002598  00000000 00000000                        1802+         DS    FD                    gap
000025A0  00000000 00000000                        1803+XCO21    DS    XL16               Cross check Output
000025A8  00000000 00000000
000025B0  00000000 00000000                        1804+         DS    XL16
000025B8  00000000 00000000
000025C0  00000000 00000000                        1805+         DS    FD                    gap
000025C8  00000000                                 1806+FPC_XC_21 DS   F                 1st cross check FPC
000025CC  00000000                                 1807+         DS    F                 2nd cross check FPC
000025D0  00000000 00000000                        1808+         DS    FD                    gap
000025D8  00000000                                 1809+         DS    F                  debug area
000025DC  00000000                                 1810+         DS    F
                                                   1811+*
000025E0                                           1812+X21      DS    0F
000025E0  B29D 852C                      0000072C  1813+         LFPC  FPCMOFF            initialize FPC
000025E4  E320 500C 0014                 0000254C  1814+         LGF   R2,V2_21           get v2
000025EA  E767 2000 0C36                 00000000  1815+         VLM   V22,V23,0(R2)
000025F0  E666 7002 0E75                           1816+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
000025F6  B29C 5040                      00002580  1817+         STFPC FPC_R_21           save FPC
000025FA  E760 5028 080E                 00002568  1818+         VST   V22,V1O21          save instruction result
00002600  07FB                                     1819+         BR    R11                return
00002604                                           1820+RE21     DS    0F                 expected 16 byte result
00002604                                           1821+         DROP  R5
00002604  7FFF0000 00000000                        1822          DC    XL16'7FFF000000000000 FFFF000000000000'
0000260C  FFFF0000 00000000
00002614  7F800000 00000000                        1823          DC    XL16'7F80000000000000 0000000000000000'
0000261C  00000000 00000000
00002624  FF800000 00000000                        1824          DC    XL16'FF80000000000000 0000000000000000'
0000262C  00000000 00000000
                                                   1825
                                                   1826 * +QNaN, -QNaN
                                                   1827          VRR_C VCRNF,0,2,80,00,S                  skip xc: lost digits
00002638                                           1828+         DS    0FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002638                      00002638            1829+          USING *,R5                base for test data and test routine
00002638  000026D8                                1830+T22       DC    A(X22)              address of test routine
0000263C  0016                                    1831+          DC    H'22'               test number
0000263E  00                                      1832+          DC    X'00'
0000263F  E2                                      1833+          DC    CL1'S'              Y = skip cross check
00002640  00                                      1834+          DC    HL1'0'              m4
00002641  02                                      1835+          DC    HL1'2'              m5
00002642  80                                      1836+FLG22     DC    X'80'               expected FPC flags
00002643  00                                      1837+VXC22     DC    X'00'               expected VXC
00002644  0000270C                                1838+V2_22     DC    A(RE22+16)          address of v2: 16-byte packed decimal
00002648  E5C3D9D5 C6404040                       1839+          DC    CL8'VCRNF'          instruction name
00002650  00000010                                1840+          DC    A(16)               result length
00002654  000026FC                                1841+          DC    A(RE22)             address of expected resul
00002658  00000000 00000000                       1842+          DS    FD                  gap
00002660  00000000 00000000                       1843+V1O22     DS    XL16                V1 output
00002668  00000000 00000000
00002670  00000000 00000000                       1844+          DS    FD                     gap
00002678  00000000                                1845+FPC_R_22 DS     F                   FPC after instruction
00002680  00000000 00000000                       1846+          DS    FD                     gap
00002688  40404040 40404040                       1847+          DC    CL8' '              was cross check skipped?
00002690  00000000 00000000                       1848+          DS    FD                     gap
00002698  00000000 00000000                       1849+XCO22     DS    XL16                Cross check Output
000026A0  00000000 00000000
000026A8  00000000 00000000                       1850+          DS    XL16
000026B0  00000000 00000000
000026B8  00000000 00000000                       1851+          DS    FD                     gap
000026C0  00000000                                1852+FPC_XC_22 DS    F                  1st cross check FPC
000026C4  00000000                                1853+          DS    F                  2nd cross check FPC
000026C8  00000000 00000000                       1854+          DS    FD                     gap
000026D0  00000000                                1855+          DS    F                   debug area
000026D4  00000000                                1856+          DS    F
                                                  1857+*
000026D8                                          1858+X22       DS    0F
000026D8  B29D 852C                     0000072C  1859+          LFPC  FPCMOFF             initialize FPC
000026DC  E320 500C 0014                00002644  1860+          LGF   R2,V2_22            get v2
000026E2  E767 2000 0C36                00000000  1861+          VLM   V22,V23,0(R2)
000026E8  E666 7002 0E75                          1862+          VCRNF V22,V22,V23,0,2         test instruction (dest is source)
000026EE  B29C 5040                     00002678  1863+          STFPC FPC_R_22            save FPC
000026F2  E760 5028 080E                00002660  1864+          VST   V22,V1O22           save instruction result
000026F8  07FB                                    1865+          BR    R11                 return
000026FC                                          1866+RE22      DS    0F                  expected 16 byte result
000026FC                                          1867+          DROP  R5
000026FC  7FFF0000 00000000                       1868           DC    XL16'7FFF000000000000 FFFF000000000000'
00002704  FFFF0000 00000000
0000270C  7FC10000 00000000                       1869           DC    XL16'7FC1000000000000 0000000000000000'
00002714  00000000 00000000
0000271C  FFC10000 00000000                       1870           DC    XL16'FFC1000000000000 0000000000000000'
00002724  00000000 00000000
                                                  1871
                                                  1872 * +SNaN, -SNaN
                                                  1873           VRR_C VCRNF,0,2,80,00,S                skip xc: lost digits
00002730                                          1874+          DS    0FD
00002730                      00002730            1875+          USING *,R5                base for test data and test routine
00002730  000027D0                                1876+T23       DC    A(X23)              address of test routine
00002734  0017                                    1877+          DC    H'23'               test number
00002736  00                                      1878+          DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002737  E2                                       1879+         DC    CL1'S'            Y = skip cross check
00002738  00                                       1880+         DC    HL1'0'            m4
00002739  02                                       1881+         DC    HL1'2'            m5
0000273A  80                                       1882+FLG23    DC    X'80'             expected FPC flags
0000273B  00                                       1883+VXC23    DC    X'00'             expected VXC
0000273C  00002804                                 1884+V2_23    DC    A(RE23+16)        address of v2: 16-byte packed decimal
00002740  E5C3D9D5 C6404040                        1885+         DC    CL8'VCRNF'        instruction name
00002748  00000010                                 1886+         DC    A(16)             result length
0000274C  000027F4                                 1887+         DC    A(RE23)           address of expected resul
00002750  00000000 00000000                        1888+         DS    FD                gap
00002758  00000000 00000000                        1889+V1O23    DS    XL16              V1 output
00002760  00000000 00000000
00002768  00000000 00000000                        1890+         DS    FD                   gap
00002770  00000000                                 1891+FPC_R_23 DS    F                 FPC after instruction
00002778  00000000 00000000                        1892+         DS    FD                   gap
00002780  40404040 40404040                        1893+         DC    CL8' '            was cross check skipped?
00002788  00000000 00000000                        1894+         DS    FD                   gap
00002790  00000000 00000000                        1895+XCO23    DS    XL16              Cross check Output
00002798  00000000 00000000
000027A0  00000000 00000000                        1896+         DS    XL16
000027A8  00000000 00000000
000027B0  00000000 00000000                        1897+         DS    FD                   gap
000027B8  00000000                                 1898+FPC_XC_23 DS   F                1st cross check FPC
000027BC  00000000                                 1899+         DS    F                2nd cross check FPC
000027C0  00000000 00000000                        1900+         DS    FD                   gap
000027C8  00000000                                 1901+         DS    F                 debug area
000027CC  00000000                                 1902+         DS    F
                                                   1903+*
000027D0                                           1904+X23      DS    0F
000027D0  B29D 852C                      0000072C  1905+         LFPC  FPCMOFF           initialize FPC
000027D4  E320 500C 0014                 0000273C  1906+         LGF   R2,V2_23          get v2
000027DA  E767 2000 0C36                 00000000  1907+         VLM   V22,V23,0(R2)
000027E0  E666 7002 0E75                           1908+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000027E6  B29C 5040                      00002770  1909+         STFPC FPC_R_23          save FPC
000027EA  E760 5028 080E                 00002758  1910+         VST   V22,V1O23         save instruction result
000027F0  07FB                                     1911+         BR    R11               return
000027F4                                           1912+RE23     DS    0F                expected 16 byte result
000027F4                                           1913+         DROP  R5
000027F4  7FFF0000 00000000                        1914          DC    XL16'7FFF000000000000 FFFF000000000000'
000027FC  FFFF0000 00000000
00002804  7F810000 00000000                        1915          DC    XL16'7F81000000000000 0000000000000000'
0000280C  00000000 00000000
00002814  FF810000 00000000                        1916          DC    XL16'FF81000000000000 0000000000000000'
0000281C  00000000 00000000
                                                   1917
                                                   1918 * Additional tests - multiple excecptions
                                                   1919 *--------------------------------------------------------------------
                                                   1920
                                                   1921 * +Subnormal, -Infinity
                                                   1922          VRR_C VCRNF,0,2,30,00,S                skip xc: lost digits
00002828                                           1923+         DS    0FD
00002828                       00002828            1924+         USING *,R5              base for test data and test routine
00002828  000028C8                                 1925+T24      DC    A(X24)            address of test routine
0000282C  0018                                     1926+         DC    H'24'             test number
0000282E  00                                       1927+         DC    X'00'
0000282F  E2                                       1928+         DC    CL1'S'            Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002830  00                                     1929+          DC    HL1'0'            m4
00002831  02                                     1930+          DC    HL1'2'            m5
00002832  30                                     1931+FLG24     DC    X'30'             expected FPC flags
00002833  00                                     1932+VXC24     DC    X'00'             expected VXC
00002834  000028FC                               1933+V2_24     DC    A(RE24+16)        address of v2: 16-byte packed decimal
00002838  E5C3D9D5 C6404040                      1934+          DC    CL8'VCRNF'        instruction name
00002840  00000010                               1935+          DC    A(16)             result length
00002844  000028EC                               1936+          DC    A(RE24)           address of expected resul
00002848  00000000 00000000                      1937+          DS    FD                gap
00002850  00000000 00000000                      1938+V1O24     DS    XL16              V1 output
00002858  00000000 00000000
00002860  00000000 00000000                      1939+          DS    FD                   gap
00002868  00000000                               1940+FPC_R_24 DS    F                 FPC after instruction
00002870  00000000 00000000                      1941+          DS    FD                   gap
00002878  40404040 40404040                      1942+          DC    CL8' '            was cross check skipped?
00002880  00000000 00000000                      1943+          DS    FD                   gap
00002888  00000000 00000000                      1944+XCO24     DS    XL16              Cross check Output
00002890  00000000 00000000
00002898  00000000 00000000                      1945+          DS    XL16
000028A0  00000000 00000000
000028A8  00000000 00000000                      1946+          DS    FD                   gap
000028B0  00000000                               1947+FPC_XC_24 DS   F                1st cross check FPC
000028B4  00000000                               1948+          DS    F                2nd cross check FPC
000028B8  00000000 00000000                      1949+          DS    FD                   gap
000028C0  00000000                               1950+          DS    F                 debug area
000028C4  00000000                               1951+          DS    F
                                                 1952+*
000028C8                                         1953+X24       DS    0F
000028C8  B29D 852C                    0000072C  1954+          LFPC  FPCMOFF           initialize FPC
000028CC  E320 500C 0014               00002834  1955+          LGF   R2,V2_24          get v2
000028D2  E767 2000 0C36               00000000  1956+          VLM   V22,V23,0(R2)
000028D8  E666 7002 0E75                         1957+          VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000028DE  B29C 5040                    00002868  1958+          STFPC FPC_R_24          save FPC
000028E2  E760 5028 080E               00002850  1959+          VST   V22,V1O24         save instruction result
000028E8  07FB                                   1960+          BR    R11               return
000028EC                                         1961+RE24      DS    0F                expected 16 byte result
000028EC                                         1962+          DROP  R5
000028EC  00000000 00000000                      1963           DC    XL16'0000000000000000 FFFF000000000000'
000028F4  FFFF0000 00000000
000028FC  007FFFFF 00000000                      1964           DC    XL16'007FFFFF00000000 0000000000000000'
00002904  00000000 00000000
0000290C  FF800000 00000000                      1965           DC    XL16'FF80000000000000 0000000000000000'
00002914  00000000 00000000
                                                 1966
                                                 1967 * -Infinity, +Subnormal
                                                 1968           VRR_C VCRNF,0,2,30,00,S                 skip xc: lost digits
00002920                                         1969+          DS    0FD
00002920                     00002920            1970+          USING *,R5              base for test data and test routine
00002920  000029C0                               1971+T25       DC    A(X25)            address of test routine
00002924  0019                                   1972+          DC    H'25'             test number
00002926  00                                     1973+          DC    X'00'
00002927  E2                                     1974+          DC    CL1'S'            Y = skip cross check
00002928  00                                     1975+          DC    HL1'0'            m4
00002929  02                                     1976+          DC    HL1'2'            m5
0000292A  30                                     1977+FLG25     DC    X'30'             expected FPC flags
0000292B  00                                     1978+VXC25     DC    X'00'             expected VXC
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

0000292C  000029F4                               1979+V2_25     DC    A(RE25+16)         address of v2: 16-byte packed decimal
00002930  E5C3D9D5 C6404040                      1980+          DC    CL8'VCRNF'         instruction name
00002938  00000010                               1981+          DC    A(16)              result length
0000293C  000029E4                               1982+          DC    A(RE25)            address of expected resul
00002940  00000000 00000000                      1983+          DS    FD                 gap
00002948  00000000 00000000                      1984+V1O25     DS    XL16               V1 output
00002950  00000000 00000000
00002958  00000000 00000000                      1985+          DS    FD                    gap
00002960  00000000                               1986+FPC_R_25  DS    F                  FPC after instruction
00002968  00000000 00000000                      1987+          DS    FD                    gap
00002970  40404040 40404040                      1988+          DC    CL8' '             was cross check skipped?
00002978  00000000 00000000                      1989+          DS    FD                    gap
00002980  00000000 00000000                      1990+XCO25     DS    XL16               Cross check Output
00002988  00000000 00000000
00002990  00000000 00000000                      1991+          DS    XL16
00002998  00000000 00000000
000029A0  00000000 00000000                      1992+          DS    FD                    gap
000029A8  00000000                               1993+FPC_XC_25 DS    F                 1st cross check FPC
000029AC  00000000                               1994+          DS    F                 2nd cross check FPC
000029B0  00000000 00000000                      1995+          DS    FD                    gap
000029B8  00000000                               1996+          DS    F                  debug area
000029BC  00000000                               1997+          DS    F
                                                 1998+*
000029C0                                         1999+X25       DS    0F
000029C0  B29D 852C                    0000072C  2000+          LFPC  FPCMOFF            initialize FPC
000029C4  E320 500C 0014               0000292C  2001+          LGF   R2,V2_25           get v2
000029CA  E767 2000 0C36               00000000  2002+          VLM   V22,V23,0(R2)
000029D0  E666 7002 0E75                         2003+          VCRNF V22,V22,V23,0,2        test instruction (dest is source)
000029D6  B29C 5040                    00002960  2004+          STFPC FPC_R_25           save FPC
000029DA  E760 5028 080E               00002948  2005+          VST   V22,V1O25          save instruction result
000029E0  07FB                                   2006+          BR    R11                return
000029E4                                         2007+RE25      DS    0F                 expected 16 byte result
000029E4                                         2008+          DROP  R5
000029E4  00000000 00000000                      2009           DC    XL16'0000000000000000 FFFF000000000000'
000029EC  FFFF0000 00000000
000029F4  00000000 00000000                      2010           DC    XL16'0000000000000000 0000000000000000'
000029FC  00000000 00000000
00002A04  FF800000 00000000                      2011           DC    XL16'FF80000000000000 007FFFFF00000000'
00002A0C  007FFFFF 00000000
                                                 2012
                                                 2013 * +Infinity, -QNaN
                                                 2014           VRR_C VCRNF,0,2,A0,00,S                skip xc: lost digits
00002A18                                         2015+          DS    0FD
00002A18                      00002A18           2016+          USING *,R5               base for test data and test routine
00002A18  00002AB8                               2017+T26       DC    A(X26)             address of test routine
00002A1C  001A                                   2018+          DC    H'26'              test number
00002A1E  00                                     2019+          DC    X'00'
00002A1F  E2                                     2020+          DC    CL1'S'             Y = skip cross check
00002A20  00                                     2021+          DC    HL1'0'             m4
00002A21  02                                     2022+          DC    HL1'2'             m5
00002A22  A0                                     2023+FLG26     DC    X'A0'              expected FPC flags
00002A23  00                                     2024+VXC26     DC    X'00'              expected VXC
00002A24  00002AEC                               2025+V2_26     DC    A(RE26+16)         address of v2: 16-byte packed decimal
00002A28  E5C3D9D5 C6404040                      2026+          DC    CL8'VCRNF'         instruction name
00002A30  00000010                               2027+          DC    A(16)              result length
00002A34  00002ADC                               2028+          DC    A(RE26)            address of expected resul
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002A38  00000000 00000000                        2029+          DS    FD                 gap
00002A40  00000000 00000000                        2030+V1O26     DS    XL16               V1 output
00002A48  00000000 00000000
00002A50  00000000 00000000                        2031+          DS    FD                    gap
00002A58  00000000                                 2032+FPC_R_26  DS    F                  FPC after instruction
00002A60  00000000 00000000                        2033+          DS    FD                    gap
00002A68  40404040 40404040                        2034+          DC    CL8' '             was cross check skipped?
00002A70  00000000 00000000                        2035+          DS    FD                    gap
00002A78  00000000 00000000                        2036+XCO26     DS    XL16               Cross check Output
00002A80  00000000 00000000
00002A88  00000000 00000000                        2037+          DS    XL16
00002A90  00000000 00000000
00002A98  00000000 00000000                        2038+          DS    FD                    gap
00002AA0  00000000                                 2039+FPC_XC_26 DS    F                 1st cross check FPC
00002AA4  00000000                                 2040+          DS    F                 2nd cross check FPC
00002AA8  00000000 00000000                        2041+          DS    FD                    gap
00002AB0  00000000                                 2042+          DS    F                  debug area
00002AB4  00000000                                 2043+          DS    F
                                                   2044+*
00002AB8                                           2045+X26       DS    0F
00002AB8  B29D 852C                      0000072C  2046+          LFPC  FPCMOFF            initialize FPC
00002ABC  E320 500C 0014                 00002A24  2047+          LGF   R2,V2_26           get v2
00002AC2  E767 2000 0C36                 00000000  2048+          VLM   V22,V23,0(R2)
00002AC8  E666 7002 0E75                           2049+          VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00002ACE  B29C 5040                      00002A58  2050+          STFPC FPC_R_26           save FPC
00002AD2  E760 5028 080E                 00002A40  2051+          VST   V22,V1O26          save instruction result
00002AD8  07FB                                     2052+          BR    R11                return
00002ADC                                           2053+RE26      DS    0F                 expected 16 byte result
00002ADC                                           2054+          DROP  R5
00002ADC  7FFF0000 00000000                        2055           DC    XL16'7FFF000000000000 FFFF000000000000'
00002AE4  FFFF0000 00000000
00002AEC  7F800000 00000000                        2056           DC    XL16'7F80000000000000 0000000000000000'
00002AF4  00000000 00000000
00002AFC  FFC10000 00000000                        2057           DC    XL16'FFC1000000000000 0000000000000000'
00002B04  00000000 00000000
                                                   2058
                                                   2059 * +Infinity, -QNaN, +Infinity
                                                   2060           VRR_C VCRNF,0,2,A0,00,S                 skip xc: lost digits
00002B10                                           2061+          DS    0FD
00002B10                       00002B10            2062+          USING *,R5               base for test data and test routine
00002B10  00002BB0                                 2063+T27       DC    A(X27)             address of test routine
00002B14  001B                                     2064+          DC    H'27'              test number
00002B16  00                                       2065+          DC    X'00'
00002B17  E2                                       2066+          DC    CL1'S'             Y = skip cross check
00002B18  00                                       2067+          DC    HL1'0'             m4
00002B19  02                                       2068+          DC    HL1'2'             m5
00002B1A  A0                                       2069+FLG27     DC    X'A0'              expected FPC flags
00002B1B  00                                       2070+VXC27     DC    X'00'              expected VXC
00002B1C  00002BE4                                 2071+V2_27     DC    A(RE27+16)         address of v2: 16-byte packed decimal
00002B20  E5C3D9D5 C6404040                        2072+          DC    CL8'VCRNF'         instruction name
00002B28  00000010                                 2073+          DC    A(16)              result length
00002B2C  00002BD4                                 2074+          DC    A(RE27)            address of expected resul
00002B30  00000000 00000000                        2075+          DS    FD                 gap
00002B38  00000000 00000000                        2076+V1O27     DS    XL16               V1 output
00002B40  00000000 00000000
00002B48  00000000 00000000                        2077+          DS    FD                    gap
```

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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002B50  00000000                                2078+FPC_R_27 DS    F                  FPC after instruction
00002B58  00000000 00000000                       2079+         DS    FD                    gap
00002B60  40404040 40404040                       2080+         DC    CL8' '             was cross check skipped?
00002B68  00000000 00000000                       2081+         DS    FD                    gap
00002B70  00000000 00000000                       2082+XCO27    DS    XL16               Cross check Output
00002B78  00000000 00000000
00002B80  00000000 00000000                       2083+         DS    XL16
00002B88  00000000 00000000
00002B90  00000000 00000000                       2084+         DS    FD                    gap
00002B98  00000000                                2085+FPC_XC_27 DS   F                 1st cross check FPC
00002B9C  00000000                                2086+         DS    F                 2nd cross check FPC
00002BA0  00000000 00000000                       2087+         DS    FD                    gap
00002BA8  00000000                                2088+         DS    F                  debug area
00002BAC  00000000                                2089+         DS    F
                                                  2090+*
00002BB0                                          2091+X27      DS    0F
00002BB0  B29D 852C                     0000072C  2092+         LFPC  FPCMOFF            initialize FPC
00002BB4  E320 500C 0014                00002B1C  2093+         LGF   R2,V2_27           get v2
00002BBA  E767 2000 0C36                00000000  2094+         VLM   V22,V23,0(R2)
00002BC0  E666 7002 0E75                          2095+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00002BC6  B29C 5040                     00002B50  2096+         STFPC FPC_R_27           save FPC
00002BCA  E760 5028 080E                00002B38  2097+         VST   V22,V1O27          save instruction result
00002BD0  07FB                                    2098+         BR    R11                return
00002BD4                                          2099+RE27     DS    0F                 expected 16 byte result
00002BD4                                          2100+         DROP  R5
00002BD4  7FFF0000 00000000                       2101          DC    XL16'7FFF000000000000 FFFF00007FFF0000'
00002BDC  FFFF0000 7FFF0000
00002BE4  7F800000 00000000                       2102          DC    XL16'7F80000000000000 0000000000000000'
00002BEC  00000000 00000000
00002BF4  FFC10000 00000000                       2103          DC    XL16'FFC1000000000000 7F80000000000000'
00002BFC  7F800000 00000000
                                                  2104
                                                  2105 * +Infinity, -QNaN, +Subnormal
                                                  2106          VRR_C VCRNF,0,2,B0,00,S                skip xc: lost digits
00002C08                                          2107+         DS    0FD
00002C08                       00002C08           2108+         USING *,R5               base for test data and test routine
00002C08  00002CA8                                2109+T28      DC    A(X28)             address of test routine
00002C0C  001C                                    2110+         DC    H'28'              test number
00002C0E  00                                      2111+         DC    X'00'
00002C0F  E2                                      2112+         DC    CL1'S'             Y = skip cross check
00002C10  00                                      2113+         DC    HL1'0'             m4
00002C11  02                                      2114+         DC    HL1'2'             m5
00002C12  B0                                      2115+FLG28    DC    X'B0'              expected FPC flags
00002C13  00                                      2116+VXC28    DC    X'00'              expected VXC
00002C14  00002CDC                                2117+V2_28    DC    A(RE28+16)         address of v2: 16-byte packed decimal
00002C18  E5C3D9D5 C6404040                       2118+         DC    CL8'VCRNF'         instruction name
00002C20  00000010                                2119+         DC    A(16)              result length
00002C24  00002CCC                                2120+         DC    A(RE28)            address of expected resul
00002C28  00000000 00000000                       2121+         DS    FD                 gap
00002C30  00000000 00000000                       2122+V1O28    DS    XL16               V1 output
00002C38  00000000 00000000
00002C40  00000000 00000000                       2123+         DS    FD                    gap
00002C48  00000000                                2124+FPC_R_28 DS    F                  FPC after instruction
00002C50  00000000 00000000                       2125+         DS    FD                    gap
00002C58  40404040 40404040                       2126+         DC    CL8' '             was cross check skipped?
00002C60  00000000 00000000                       2127+         DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002C68  00000000 00000000                      2128+XCO28    DS    XL16              Cross check Output
00002C70  00000000 00000000
00002C78  00000000 00000000                      2129+         DS    XL16
00002C80  00000000 00000000
00002C88  00000000 00000000                      2130+         DS    FD                    gap
00002C90  00000000                               2131+FPC_XC_28 DS   F                1st cross check FPC
00002C94  00000000                               2132+         DS    F                2nd cross check FPC
00002C98  00000000 00000000                      2133+         DS    FD                    gap
00002CA0  00000000                               2134+         DS    F                 debug area
00002CA4  00000000                               2135+         DS    F
                                                 2136+*
00002CA8                                         2137+X28      DS    0F
00002CA8  B29D 852C                    0000072C  2138+         LFPC  FPCMOFF           initialize FPC
00002CAC  E320 500C 0014               00002C14  2139+         LGF   R2,V2_28          get v2
00002CB2  E767 2000 0C36               00000000  2140+         VLM   V22,V23,0(R2)
00002CB8  E666 7002 0E75                         2141+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002CBE  B29C 5040                    00002C48  2142+         STFPC FPC_R_28          save FPC
00002CC2  E760 5028 080E               00002C30  2143+         VST   V22,V1O28         save instruction result
00002CC8  07FB                                   2144+         BR    R11               return
00002CCC                                         2145+RE28     DS    0F                expected 16 byte result
00002CCC                                         2146+         DROP  R5
00002CCC  7FFF0000 00000000                      2147          DC    XL16'7FFF000000000000 FFFF000000000000'
00002CD4  FFFF0000 00000000
00002CDC  7F800000 00000000                      2148          DC    XL16'7F80000000000000 0000000000000000'
00002CE4  00000000 00000000
00002CEC  FFC10000 00000000                      2149          DC    XL16'FFC1000000000000 007FFFFF00000000'
00002CF4  007FFFFF 00000000
                                                 2150
                                                 2151
00002CFC  00000000                               2152          DC    F'0'     END OF TABLE
00002D00  00000000                               2153          DC    F'0'
                                                 2154 *
                                                 2155 * table of pointers to individual tests
                                                 2156 *
00002D04                                         2157 E6TESTS  DS    0F
                                                 2158          PTTABLE
00002D04                                         2159+TTABLE   DS    0F
00002D04  000011E0                               2160+         DC    A(T1)             TEST &CUR
00002D08  000012D8                               2161+         DC    A(T2)             TEST &CUR
00002D0C  000013D0                               2162+         DC    A(T3)             TEST &CUR
00002D10  000014C8                               2163+         DC    A(T4)             TEST &CUR
00002D14  000015C0                               2164+         DC    A(T5)             TEST &CUR
00002D18  000016B8                               2165+         DC    A(T6)             TEST &CUR
00002D1C  000017B0                               2166+         DC    A(T7)             TEST &CUR
00002D20  000018A8                               2167+         DC    A(T8)             TEST &CUR
00002D24  000019A0                               2168+         DC    A(T9)             TEST &CUR
00002D28  00001A98                               2169+         DC    A(T10)            TEST &CUR
00002D2C  00001B90                               2170+         DC    A(T11)            TEST &CUR
00002D30  00001C88                               2171+         DC    A(T12)            TEST &CUR
00002D34  00001D80                               2172+         DC    A(T13)            TEST &CUR
00002D38  00001E78                               2173+         DC    A(T14)            TEST &CUR
00002D3C  00001F70                               2174+         DC    A(T15)            TEST &CUR
00002D40  00002068                               2175+         DC    A(T16)            TEST &CUR
00002D44  00002160                               2176+         DC    A(T17)            TEST &CUR
00002D48  00002258                               2177+         DC    A(T18)            TEST &CUR
00002D4C  00002350                               2178+         DC    A(T19)            TEST &CUR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002D50  00002448                                2179+          DC    A(T20)            TEST &CUR
00002D54  00002540                                2180+          DC    A(T21)            TEST &CUR
00002D58  00002638                                2181+          DC    A(T22)            TEST &CUR
00002D5C  00002730                                2182+          DC    A(T23)            TEST &CUR
00002D60  00002828                                2183+          DC    A(T24)            TEST &CUR
00002D64  00002920                                2184+          DC    A(T25)            TEST &CUR
00002D68  00002A18                                2185+          DC    A(T26)            TEST &CUR
00002D6C  00002B10                                2186+          DC    A(T27)            TEST &CUR
00002D70  00002C08                                2187+          DC    A(T28)            TEST &CUR
                                                  2188+*
00002D74  00000000                                2189+          DC    A(0)              END OF TABLE
00002D78  00000000                                2190+          DC    A(0)
                                                  2191
00002D7C  00000000                                2192           DC    F'0'     END OF TABLE
00002D80  00000000                                2193           DC    F'0'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  2195 ******************************************************************
                                                  2196 *        Register equates
                                                  2197 ******************************************************************

                                00000000  00000001  2199 R0       EQU   0
                                00000001  00000001  2200 R1       EQU   1
                                00000002  00000001  2201 R2       EQU   2
                                00000003  00000001  2202 R3       EQU   3
                                00000004  00000001  2203 R4       EQU   4
                                00000005  00000001  2204 R5       EQU   5
                                00000006  00000001  2205 R6       EQU   6
                                00000007  00000001  2206 R7       EQU   7
                                00000008  00000001  2207 R8       EQU   8
                                00000009  00000001  2208 R9       EQU   9
                                0000000A  00000001  2209 R10      EQU   10
                                0000000B  00000001  2210 R11      EQU   11
                                0000000C  00000001  2211 R12      EQU   12
                                0000000D  00000001  2212 R13      EQU   13
                                0000000E  00000001  2213 R14      EQU   14
                                0000000F  00000001  2214 R15      EQU   15

                                                  2216 ******************************************************************
                                                  2217 *        Register equates
                                                  2218 ******************************************************************

                                00000000  00000001  2220 FPR0     EQU   0
                                00000001  00000001  2221 FPR1     EQU   1
                                00000002  00000001  2222 FPR2     EQU   2
                                00000003  00000001  2223 FPR3     EQU   3
                                00000004  00000001  2224 FPR4     EQU   4
                                00000005  00000001  2225 FPR5     EQU   5
                                00000006  00000001  2226 FPR6     EQU   6
                                00000007  00000001  2227 FPR7     EQU   7
                                00000008  00000001  2228 FPR8     EQU   8
                                00000009  00000001  2229 FPR9     EQU   9
                                0000000A  00000001  2230 FPR10    EQU   10
                                0000000B  00000001  2231 FPR11    EQU   11
                                0000000C  00000001  2232 FPR12    EQU   12
                                0000000D  00000001  2233 FPR13    EQU   13
                                0000000E  00000001  2234 FPR14    EQU   14
                                0000000F  00000001  2235 FPR15    EQU   15

                                                  2237 ******************************************************************
                                                  2238 *        Register equates
                                                  2239 ******************************************************************

                                00000000  00000001  2241 V0       EQU   0
                                00000001  00000001  2242 V1       EQU   1
                                00000002  00000001  2243 V2       EQU   2
                                00000003  00000001  2244 V3       EQU   3
                                00000004  00000001  2245 V4       EQU   4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                              00000005  00000001  2246 V5          EQU   5
                              00000006  00000001  2247 V6          EQU   6
                              00000007  00000001  2248 V7          EQU   7
                              00000008  00000001  2249 V8          EQU   8
                              00000009  00000001  2250 V9          EQU   9
                              0000000A  00000001  2251 V10         EQU   10
                              0000000B  00000001  2252 V11         EQU   11
                              0000000C  00000001  2253 V12         EQU   12
                              0000000D  00000001  2254 V13         EQU   13
                              0000000E  00000001  2255 V14         EQU   14
                              0000000F  00000001  2256 V15         EQU   15
                              00000010  00000001  2257 V16         EQU   16
                              00000011  00000001  2258 V17         EQU   17
                              00000012  00000001  2259 V18         EQU   18
                              00000013  00000001  2260 V19         EQU   19
                              00000014  00000001  2261 V20         EQU   20
                              00000015  00000001  2262 V21         EQU   21
                              00000016  00000001  2263 V22         EQU   22
                              00000017  00000001  2264 V23         EQU   23
                              00000018  00000001  2265 V24         EQU   24
                              00000019  00000001  2266 V25         EQU   25
                              0000001A  00000001  2267 V26         EQU   26
                              0000001B  00000001  2268 V27         EQU   27
                              0000001C  00000001  2269 V28         EQU   28
                              0000001D  00000001  2270 V29         EQU   29
                              0000001E  00000001  2271 V30         EQU   30
                              0000001F  00000001  2272 V31         EQU   31
                                                  2273
                                                  2274             END
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
BEGIN	I	00000200	2	172	136 168 169 170
CHECKRES	U	00000324	1	252	249
CHECKVXC	U	00000316	1	247	244
CTLR0	F	00000724	4	570	182 183 184 185
DECNUM	C	00001190	16	696	326 328 335 337 342 344 373 375 382 384 389 391 409 411 418 420 425 427 446 448 455 457 462 464
DONEXT	U	00000346	1	266	260
E6TEST	4	00000000	160	710	231
E6TESTS	F	00002D04	4	2157	224
EDIT	X	00001164	18	691	327 336 343 374 383 390 410 419 426 447 456 463
ENDTEST	U	000005F4	1	484	229
EOJ	I	00000708	4	560	217 487
EOJPSW	D	000006F8	8	558	560
FAILCONT	U	000005E4	1	474	
FAILED	F	00001000	4	611	323 354 370 406 443 476 485
FAILMSG	U	00000574	1	441	264
FAILPSW	D	00000710	8	562	564
FAILTEST	I	00000720	4	564	488
FB0001	F	00000298	8	201	205 206 208
FLG	X	0000000A	1	717	243 381
FLG1	X	000011EA	1	868	
FLG10	X	00001AA2	1	1282	
FLG11	X	00001B9A	1	1328	
FLG12	X	00001C92	1	1374	
FLG13	X	00001D8A	1	1420	
FLG14	X	00001E82	1	1468	
FLG15	X	00001F7A	1	1514	
FLG16	X	00002072	1	1560	
FLG17	X	0000216A	1	1606	
FLG18	X	00002262	1	1652	
FLG19	X	0000235A	1	1698	
FLG2	X	000012E2	1	914	
FLG20	X	00002452	1	1744	
FLG21	X	0000254A	1	1790	
FLG22	X	00002642	1	1836	
FLG23	X	0000273A	1	1882	
FLG24	X	00002832	1	1931	
FLG25	X	0000292A	1	1977	
FLG26	X	00002A22	1	2023	
FLG27	X	00002B1A	1	2069	
FLG28	X	00002C12	1	2115	
FLG3	X	000013DA	1	960	
FLG4	X	000014D2	1	1006	
FLG5	X	000015CA	1	1052	
FLG6	X	000016C2	1	1098	
FLG7	X	000017BA	1	1144	
FLG8	X	000018B2	1	1190	
FLG9	X	000019AA	1	1236	
FPCMIE	X	00000740	4	577	
FPCMIO	X	00000734	4	574	
FPCMOFF	X	0000072C	4	572	294 301 891 937 983 1029 1075 1121 1167 1213 1259 1305 1351 1397 1443 1491 1537 1583 1629 1675 1721 1767 1813 1859 1905 1954 2000 2046 2092 2138
FPCMON	X	00000730	4	573	
FPCMOV	X	00000738	4	575	
FPCMSG	U	00000490	1	368	245

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES					
FPCMUN	X	0000073C	4	576						
FPCPEXP	C	00001052	3	626	385					
FPCPGOT	C	00001067	3	629	392					
FPCPLINE	C	00001008	13	621	631	395				
FPCPLNG	U	00000063	1	631	394					
FPCPNAME	C	00001039	8	624	378					
FPCPNUM	C	00001015	3	622	376					
FPC_R	F	00000040	4	726	243	248	257	278	388	424
FPC_R_1	F	00001220	4	877	895					
FPC_R_10	F	00001AD8	4	1291	1309					
FPC_R_11	F	00001BD0	4	1337	1355					
FPC_R_12	F	00001CC8	4	1383	1401					
FPC_R_13	F	00001DC0	4	1429	1447					
FPC_R_14	F	00001EB8	4	1477	1495					
FPC_R_15	F	00001FB0	4	1523	1541					
FPC_R_16	F	000020A8	4	1569	1587					
FPC_R_17	F	000021A0	4	1615	1633					
FPC_R_18	F	00002298	4	1661	1679					
FPC_R_19	F	00002390	4	1707	1725					
FPC_R_2	F	00001318	4	923	941					
FPC_R_20	F	00002488	4	1753	1771					
FPC_R_21	F	00002580	4	1799	1817					
FPC_R_22	F	00002678	4	1845	1863					
FPC_R_23	F	00002770	4	1891	1909					
FPC_R_24	F	00002868	4	1940	1958					
FPC_R_25	F	00002960	4	1986	2004					
FPC_R_26	F	00002A58	4	2032	2050					
FPC_R_27	F	00002B50	4	2078	2096					
FPC_R_28	F	00002C48	4	2124	2142					
FPC_R_3	F	00001410	4	969	987					
FPC_R_4	F	00001508	4	1015	1033					
FPC_R_5	F	00001600	4	1061	1079					
FPC_R_6	F	000016F8	4	1107	1125					
FPC_R_7	F	000017F0	4	1153	1171					
FPC_R_8	F	000018E8	4	1199	1217					
FPC_R_9	F	000019E0	4	1245	1263					
FPC_XC1	F	00000088	4	733	298	308				
FPC_XC2	F	0000008C	4	734	305	311				
FPC_XC_1	F	00001268	4	884						
FPC_XC_10	F	00001B20	4	1298						
FPC_XC_11	F	00001C18	4	1344						
FPC_XC_12	F	00001D10	4	1390						
FPC_XC_13	F	00001E08	4	1436						
FPC_XC_14	F	00001F00	4	1484						
FPC_XC_15	F	00001FF8	4	1530						
FPC_XC_16	F	000020F0	4	1576						
FPC_XC_17	F	000021E8	4	1622						
FPC_XC_18	F	000022E0	4	1668						
FPC_XC_19	F	000023D8	4	1714						
FPC_XC_2	F	00001360	4	930						
FPC_XC_20	F	000024D0	4	1760						
FPC_XC_21	F	000025C8	4	1806						
FPC_XC_22	F	000026C0	4	1852						
FPC_XC_23	F	000027B8	4	1898						
FPC_XC_24	F	000028B0	4	1947						
FPC_XC_25	F	000029A8	4	1993						

```
     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

FPC_XC_26            F    00002AA0          4  2039
FPC_XC_27            F    00002B98          4  2085
FPC_XC_28            F    00002C90          4  2131
FPC_XC_3             F    00001458          4   976
FPC_XC_4             F    00001550          4  1022
FPC_XC_5             F    00001648          4  1068
FPC_XC_6             F    00001740          4  1114
FPC_XC_7             F    00001838          4  1160
FPC_XC_8             F    00001930          4  1206
FPC_XC_9             F    00001A28          4  1252
FPR0                 U    00000000          1  2220
FPR1                 U    00000001          1  2221
FPR10                U    0000000A          1  2230
FPR11                U    0000000B          1  2231
FPR12                U    0000000C          1  2232
FPR13                U    0000000D          1  2233
FPR14                U    0000000E          1  2234
FPR15                U    0000000F          1  2235
FPR2                 U    00000002          1  2222
FPR3                 U    00000003          1  2223
FPR4                 U    00000004          1  2224
FPR5                 U    00000005          1  2225
FPR6                 U    00000006          1  2226
FPR7                 U    00000007          1  2227
FPR8                 U    00000008          1  2228
FPR9                 U    00000009          1  2229
IMAGE                1    00000000      11652     0
K                    U    00000400          1   594   595   596   597
K64                  U    00010000          1   596
M4                   U    00000008          1   715   334   454
M5                   U    00000009          1   716   341   461
MB                   U    00100000          1   597
MSG                  I    00000638          4   520   216   503
MSGCMD               C    00000686          9   550   533   534
MSGMSG               C    0000068F        100   551   527   548   525
MSGMVC               I    00000680          6   548   531
MSGOK                I    0000064E          2   529   526
MSGRET               I    0000066E          4   544   537   540
MSGSAVE              F    00000674          4   547   523   544
NEXTE6               U    000002EC          1   226   268   479
OPNAME               C    00000010          8   720   284   331   378   414   451
PAGE                 U    00001000          1   595
PRT3                 C    0000117A         18   694   327   328   329   336   337   338   343   344   345   374   375   376   383
                                                      384   385   390   391   392   410   411   412   419   420   421   426   427
                                                      428   447   448   449   456   457   458   463   464   465
PRTLINE              C    000010CE         13   659   669   468
PRTLNG               U    00000048          1   669   467
PRTM4                C    00001107          2   664   458
PRTM5                C    00001113          2   667   465
PRTNAME              C    000010F6          8   662   451
PRTNUM               C    000010DB          3   660   449
R0                   U    00000000          1  2199   130   182   185   205   207   208   209   214   233   234   322   323   348
                                                      353   354   369   370   394   405   406   430   442   443   467   475   476
                                                      502   504   520   523   525   527   529   544
R1                   U    00000001          1  2200   215   256   257   258   259   262   263   314   315   349   395   431   468
                                                      485   486   534   548
```

```
     SYMBOL        TYPE   VALUE       LENGTH    DEFN  REFERENCES

R10                 U    0000000A          1  2209   170   179   180
R11                 U    0000000B          1  2210   236   237   897   943   989  1035  1081  1127  1173  1219  1265  1311  1357
                                                    1403  1449  1497  1543  1589  1635  1681  1727  1773  1819  1865  1911  1960
                                                    2006  2052  2098  2144
R12                 U    0000000C          1  2211   224   227   267   478
R13                 U    0000000D          1  2212
R14                 U    0000000E          1  2213
R15                 U    0000000F          1  2214   239   277   279   289   309   312   317   347   350   351   355   396   432
                                                     469   497   507   508
R2                  U    00000002          1  2201   216   325   326   333   334   335   340   341   342   372   373   380   381
                                                     382   387   388   389   408   409   416   417   418   423   424   425   445
                                                     446   453   454   455   460   461   462   502   503   504   521   523   529
                                                     530   531   533   539   544   545   892   893   938   939   984   985  1030
                                                    1031  1076  1077  1122  1123  1168  1169  1214  1215  1260  1261  1306  1307
                                                    1352  1353  1398  1399  1444  1445  1492  1493  1538  1539  1584  1585  1630
                                                    1631  1676  1677  1722  1723  1768  1769  1814  1815  1860  1861  1906  1907
                                                    1955  1956  2001  2002  2047  2048  2093  2094  2139  2140
R3                  U    00000003          1  2202
R4                  U    00000004          1  2203
R5                  U    00000005          1  2204   227   228   231   498   506   861   899   907   945   953   991   999  1037
                                                    1045  1083  1091  1129  1137  1175  1183  1221  1229  1267  1275  1313  1321
                                                    1359  1367  1405  1413  1451  1461  1499  1507  1545  1553  1591  1599  1637
                                                    1645  1683  1691  1729  1737  1775  1783  1821  1829  1867  1875  1913  1924
                                                    1962  1970  2008  2016  2054  2062  2100  2108  2146
R6                  U    00000006          1  2205
R7                  U    00000007          1  2206   245   250   397   433
R8                  U    00000008          1  2207   168   172   173   174   176
R9                  U    00000009          1  2208   169   176   177   179
RE1                 F    000012A4          4   898   870   873
RE10                F    00001B5C          4  1312  1284  1287
RE11                F    00001C54          4  1358  1330  1333
RE12                F    00001D4C          4  1404  1376  1379
RE13                F    00001E44          4  1450  1422  1425
RE14                F    00001F3C          4  1498  1470  1473
RE15                F    00002034          4  1544  1516  1519
RE16                F    0000212C          4  1590  1562  1565
RE17                F    00002224          4  1636  1608  1611
RE18                F    0000231C          4  1682  1654  1657
RE19                F    00002414          4  1728  1700  1703
RE2                 F    0000139C          4   944   916   919
RE20                F    0000250C          4  1774  1746  1749
RE21                F    00002604          4  1820  1792  1795
RE22                F    000026FC          4  1866  1838  1841
RE23                F    000027F4          4  1912  1884  1887
RE24                F    000028EC          4  1961  1933  1936
RE25                F    000029E4          4  2007  1979  1982
RE26                F    00002ADC          4  2053  2025  2028
RE27                F    00002BD4          4  2099  2071  2074
RE28                F    00002CCC          4  2145  2117  2120
RE3                 F    00001494          4   990   962   965
RE4                 F    0000158C          4  1036  1008  1011
RE5                 F    00001684          4  1082  1054  1057
RE6                 F    0000177C          4  1128  1100  1103
RE7                 F    00001874          4  1174  1146  1149
RE8                 F    0000196C          4  1220  1192  1195
RE9                 F    00001A64          4  1266  1238  1241
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
READDR	A	0000001C	4	722	262				
REG2LOW	U	000000DD	1	601					
REG2PATT	U	AABBCCDD	1	600					
RELEN	A	00000018	4	721					
RPTDWSAV	D	00000628	8	513	502	504			
RPTERROR	I	00000602	4	497	350	396	432	469	
RPTSAVE	F	00000620	4	510	497	507			
RPTSVR5	F	00000624	4	511	498	506			
SKIPXC	C	00000050	8	728	275	283	288		
SKL0001	U	0000005F	1	198	214				
SKT0001	C	0000022A	26	195	198	215			
SVOLDPSW	U	00000140	0	132					
T1	A	000011E0	4	862	2160				
T10	A	00001A98	4	1276	2169				
T11	A	00001B90	4	1322	2170				
T12	A	00001C88	4	1368	2171				
T13	A	00001D80	4	1414	2172				
T14	A	00001E78	4	1462	2173				
T15	A	00001F70	4	1508	2174				
T16	A	00002068	4	1554	2175				
T17	A	00002160	4	1600	2176				
T18	A	00002258	4	1646	2177				
T19	A	00002350	4	1692	2178				
T2	A	000012D8	4	908	2161				
T20	A	00002448	4	1738	2179				
T21	A	00002540	4	1784	2180				
T22	A	00002638	4	1830	2181				
T23	A	00002730	4	1876	2182				
T24	A	00002828	4	1925	2183				
T25	A	00002920	4	1971	2184				
T26	A	00002A18	4	2017	2185				
T27	A	00002B10	4	2063	2186				
T28	A	00002C08	4	2109	2187				
T3	A	000013D0	4	954	2162				
T4	A	000014C8	4	1000	2163				
T5	A	000015C0	4	1046	2164				
T6	A	000016B8	4	1092	2165				
T7	A	000017B0	4	1138	2166				
T8	A	000018A8	4	1184	2167				
T9	A	000019A0	4	1230	2168				
TESTING	F	00001004	4	612	234				
TNUM	H	00000004	2	712	233	325	372	408	445
TSUB	A	00000000	4	711	236				
TTABLE	F	00002D04	4	2159					
V0	U	00000000	1	2241					
V1	U	00000001	1	2242					
V10	U	0000000A	1	2251					
V11	U	0000000B	1	2252					
V12	U	0000000C	1	2253					
V13	U	0000000D	1	2254					
V14	U	0000000E	1	2255					
V15	U	0000000F	1	2256	297	299	304	306	
V16	U	00000010	1	2257	296	297	303	304	
V17	U	00000011	1	2258					
V18	U	00000012	1	2259					
V19	U	00000013	1	2260					

```
     SYMBOL        TYPE   VALUE       LENGTH    DEFN  REFERENCES

V1FUDGE              X    000011B0        16     702
V1INPUT              X    000011C0        16     703
V1O1                 X    00001208        16     875   896
V1O10                X    00001AC0        16    1289  1310
V1O11                X    00001BB8        16    1335  1356
V1O12                X    00001CB0        16    1381  1402
V1O13                X    00001DA8        16    1427  1448
V1O14                X    00001EA0        16    1475  1496
V1O15                X    00001F98        16    1521  1542
V1O16                X    00002090        16    1567  1588
V1O17                X    00002188        16    1613  1634
V1O18                X    00002280        16    1659  1680
V1O19                X    00002378        16    1705  1726
V1O2                 X    00001300        16     921   942
V1O20                X    00002470        16    1751  1772
V1O21                X    00002568        16    1797  1818
V1O22                X    00002660        16    1843  1864
V1O23                X    00002758        16    1889  1910
V1O24                X    00002850        16    1938  1959
V1O25                X    00002948        16    1984  2005
V1O26                X    00002A40        16    2030  2051
V1O27                X    00002B38        16    2076  2097
V1O28                X    00002C30        16    2122  2143
V1O3                 X    000013F8        16     967   988
V1O4                 X    000014F0        16    1013  1034
V1O5                 X    000015E8        16    1059  1080
V1O6                 X    000016E0        16    1105  1126
V1O7                 X    000017D8        16    1151  1172
V1O8                 X    000018D0        16    1197  1218
V1O9                 X    000019C8        16    1243  1264
V1OUTPUT             X    00000028        16     724   263   296   303
V2                   U    00000002         1    2243
V20                  U    00000014         1    2261
V21                  U    00000015         1    2262
V22                  U    00000016         1    2263   893   894   896   939   940   942   985   986   988  1031  1032  1034  1077
                                                      1078  1080  1123  1124  1126  1169  1170  1172  1215  1216  1218  1261  1262
                                                      1264  1307  1308  1310  1353  1354  1356  1399  1400  1402  1445  1446  1448
                                                      1493  1494  1496  1539  1540  1542  1585  1586  1588  1631  1632  1634  1677
                                                      1678  1680  1723  1724  1726  1769  1770  1772  1815  1816  1818  1861  1862
                                                      1864  1907  1908  1910  1956  1957  1959  2002  2003  2005  2048  2049  2051
                                                      2094  2095  2097  2140  2141  2143
V23                  U    00000017         1    2264   893   894   939   940   985   986  1031  1032  1077  1078  1123  1124  1169
                                                      1170  1215  1216  1261  1262  1307  1308  1353  1354  1399  1400  1445  1446
                                                      1493  1494  1539  1540  1585  1586  1631  1632  1677  1678  1723  1724  1769
                                                      1770  1815  1816  1861  1862  1907  1908  1956  1957  2002  2003  2048  2049
                                                      2094  2095  2140  2141
V24                  U    00000018         1    2265
V25                  U    00000019         1    2266
V26                  U    0000001A         1    2267
V27                  U    0000001B         1    2268
V28                  U    0000001C         1    2269
V29                  U    0000001D         1    2270
V2ADDR               A    0000000C         4     719   314
V2_1                 A    000011EC         4     870   892
V2_10                A    00001AA4         4    1284  1306
V2_11                A    00001B9C         4    1330  1352
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V2_12	A	00001C94	4	1376	1398
V2_13	A	00001D8C	4	1422	1444
V2_14	A	00001E84	4	1470	1492
V2_15	A	00001F7C	4	1516	1538
V2_16	A	00002074	4	1562	1584
V2_17	A	0000216C	4	1608	1630
V2_18	A	00002264	4	1654	1676
V2_19	A	0000235C	4	1700	1722
V2_2	A	000012E4	4	916	938
V2_20	A	00002454	4	1746	1768
V2_21	A	0000254C	4	1792	1814
V2_22	A	00002644	4	1838	1860
V2_23	A	0000273C	4	1884	1906
V2_24	A	00002834	4	1933	1955
V2_25	A	0000292C	4	1979	2001
V2_26	A	00002A24	4	2025	2047
V2_27	A	00002B1C	4	2071	2093
V2_28	A	00002C14	4	2117	2139
V2_3	A	000013DC	4	962	984
V2_4	A	000014D4	4	1008	1030
V2_5	A	000015CC	4	1054	1076
V2_6	A	000016C4	4	1100	1122
V2_7	A	000017BC	4	1146	1168
V2_8	A	000018B4	4	1192	1214
V2_9	A	000019AC	4	1238	1260
V3	U	00000003	1	2244	
V30	U	0000001E	1	2271	
V31	U	0000001F	1	2272	
V4	U	00000004	1	2245	
V5	U	00000005	1	2246	
V6	U	00000006	1	2247	
V7	U	00000007	1	2248	
V8	U	00000008	1	2249	
V9	U	00000009	1	2250	
VXC	X	0000000B	1	718	248 417
VXC1	X	000011EB	1	869	
VXC10	X	00001AA3	1	1283	
VXC11	X	00001B9B	1	1329	
VXC12	X	00001C93	1	1375	
VXC13	X	00001D8B	1	1421	
VXC14	X	00001E83	1	1469	
VXC15	X	00001F7B	1	1515	
VXC16	X	00002073	1	1561	
VXC17	X	0000216B	1	1607	
VXC18	X	00002263	1	1653	
VXC19	X	0000235B	1	1699	
VXC2	X	000012E3	1	915	
VXC20	X	00002453	1	1745	
VXC21	X	0000254B	1	1791	
VXC22	X	00002643	1	1837	
VXC23	X	0000273B	1	1883	
VXC24	X	00002833	1	1932	
VXC25	X	0000292B	1	1978	
VXC26	X	00002A23	1	2024	
VXC27	X	00002B1B	1	2070	
VXC28	X	00002C13	1	2116	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
VXC3	X	000013DB	1	961		
VXC4	X	000014D3	1	1007		
VXC5	X	000015CB	1	1053		
VXC6	X	000016C3	1	1099		
VXC7	X	000017BB	1	1145		
VXC8	X	000018B3	1	1191		
VXC9	X	000019AB	1	1237		
VXCMSG	U	00000502	1	404	250	
VXCPEXP	C	000010B5	3	645	421	
VXCPGOT	C	000010CA	3	648	428	
VXCPLINE	C	0000106B	13	640	650	431
VXCPLNG	U	00000063	1	650	430	
VXCPNAME	C	0000109C	8	643	414	
VXCPNUM	C	00001078	3	641	412	
WK1	F	00000098	4	736		
WK2	F	0000009C	4	737		
X0001	U	000002C0	1	204	192	205
X1	F	00001280	4	890	862	
X10	F	00001B38	4	1304	1276	
X11	F	00001C30	4	1350	1322	
X12	F	00001D28	4	1396	1368	
X13	F	00001E20	4	1442	1414	
X14	F	00001F18	4	1490	1462	
X15	F	00002010	4	1536	1508	
X16	F	00002108	4	1582	1554	
X17	F	00002200	4	1628	1600	
X18	F	000022F8	4	1674	1646	
X19	F	000023F0	4	1720	1692	
X2	F	00001378	4	936	908	
X20	F	000024E8	4	1766	1738	
X21	F	000025E0	4	1812	1784	
X22	F	000026D8	4	1858	1830	
X23	F	000027D0	4	1904	1876	
X24	F	000028C8	4	1953	1925	
X25	F	000029C0	4	1999	1971	
X26	F	00002AB8	4	2045	2017	
X27	F	00002BB0	4	2091	2063	
X28	F	00002CA8	4	2137	2109	
X3	F	00001470	4	982	954	
X4	F	00001568	4	1028	1000	
X5	F	00001660	4	1074	1046	
X6	F	00001758	4	1120	1092	
X7	F	00001850	4	1166	1138	
X8	F	00001948	4	1212	1184	
X9	F	00001A40	4	1258	1230	
XC0001	U	000002E8	1	218	210	
XCFAILMSG	H	000003D6	2	321	316	
XCHECK	U	0000034E	1	274	239	
XCO1	X	00001240	16	881		
XCO10	X	00001AF8	16	1295		
XCO11	X	00001BF0	16	1341		
XCO12	X	00001CE8	16	1387		
XCO13	X	00001DE0	16	1433		
XCO14	X	00001ED8	16	1481		
XCO15	X	00001FD0	16	1527		
XCO16	X	000020C8	16	1573		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES					
XC017	X	000021C0	16	1619						
XC018	X	000022B8	16	1665						
XC019	X	000023B0	16	1711						
XC02	X	00001338	16	927						
XC020	X	000024A8	16	1757						
XC021	X	000025A0	16	1803						
XC022	X	00002698	16	1849						
XC023	X	00002790	16	1895						
XC024	X	00002888	16	1944						
XC025	X	00002980	16	1990						
XC026	X	00002A78	16	2036						
XC027	X	00002B70	16	2082						
XC028	X	00002C68	16	2128						
XC03	X	00001430	16	973						
XC04	X	00001528	16	1019						
XC05	X	00001620	16	1065						
XC06	X	00001718	16	1111						
XC07	X	00001810	16	1157						
XC08	X	00001908	16	1203						
XC09	X	00001A00	16	1249						
XCOUTPUT	X	00000060	16	730	299	306	315			
XCPLINE	C	00001116	13	677	686	349				
XCPLNG	U	0000004E	1	686	348					
XCPM4	C	00001156	2	682	338					
XCPM5	C	00001161	2	684	345					
XCPNAME	C	00001145	8	680	331					
XCPTNUM	C	00001123	3	678	329					
XCR15	F	00000488	8	361	347	351				
XCRESULT	X	00000458	16	358						
XCSKIP	C	00000007	1	714	276					
XCV1	X	00000468	16	359						
XCV2	X	00000478	16	360						
XCVCRNF	F	00000380	4	293	285					
ZVE6TST	J	00000000	11652	129	132	134	138	142	610	130
=A(E6TESTS)	A	00000764	4	585	224					
=AL2(L'MSGMSG)	R	00000772	2	589	525					
=CL1'S'	C	00000774	1	590	276					
=CL8' '	C	00000750	8	582	283					
=CL8'SKIP XC '	C	00000748	8	581	275	288				
=CL8'VCRNF'	C	00000758	8	583	284					
=F'1'	F	0000076C	4	587	322	353	369	405	442	475
=F'4'	F	00000760	4	584	209					
=H'0'	H	00000770	2	588	520					
=XL4'00000008'	X	00000768	4	586	258					

MACRO	DEFN	REFERENCES															
FCHECK	81	191															
PTTABLE	817	2158															
VRR_C	753	859	905	951	997	1043	1089	1135	1181	1227	1273	1319	1365	1411	1459	1505	1551	1597
		1643	1689	1735	1781	1827	1873	1922	1968	2014	2060	2106						

DESC	SYMBOL	SIZE	POS	ADDR
Entry: 0				
Image	IMAGE	11652	0000-2D83	0000-2D83
Region		11652	0000-2D83	0000-2D83
CSECT	ZVE6TST	11652	0000-2D83	0000-2D83

```
   STMT                         FILE NAME

1     /devstor/sharedvfp/tests/zvector-e6-21-VCRNF.asm

** NO ERRORS FOUND **
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